Anthropometric - An Algo Approach



PROJECT REPORT

Submitted by

Sanath S. Shetty (1BM23CS297)
Suhas M S (1BM23CS346)
Sumanth S. Shetty (1BM23CS348)
Thilak K (1WA23CS021)

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Under the Guidance of

Dr. Jyothi S. Nayak
Professor, Department of CSE, BMSCE



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Certificate

This is to certify that the project report entitled “Anthropometric - An Algo Approach” is a bonafide work carried out by Sanath S. Shetty (USN: 1BM23CS297), Suhas M S (USN: 1BM23CS346), Sumanth S. Shetty (USN: 1BM23CS348) and Thilak K (USN: 1WA23CS021) in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science and Engineering, under my supervision during the academic year 2026–2027.

Guide Signature	HoD signature	Principal Signature
Dr. Jyothi S. Nayak	Dr. Kavitha Sooda	Dr. Bheemsha Arya

Name and signature of Examiners along with date

Examiner 1

Examiner 2

Abstract

Chest, waist and hip girths support garment sizing, ergonomics and health screening, but tape measurement needs a trained person and 3D scanners need dedicated hardware. Learning-based photographic estimators typically require large labelled training sets and upload full-body photographs to a server. This project presents a two-view pipeline that estimates body measurements from one frontal and one profile photograph and the stature, without training any model on measurement labels.

Pretrained pose and segmentation models run in the user's browser, so in the normal workflow only binary silhouettes, ten landmarks per view and the stature are transmitted; the server rejects any non-binary image. The silhouettes are aligned by weighted Procrustes analysis, scaled by stature and intersected into a conventional visual hull and a profile-guided reconstruction that caps a filtered profile depth by the vertical thickness of each frontal silhouette column, limiting the depth horizontal limbs inherit. Cross-sections are measured by separate geometric estimators, and every value records its reconstruction source, level and model.

In practice, a user opens the web page on a phone or laptop, takes or selects a photograph from the front and one from the side while standing with the arms held out, and types in their height. Within a few seconds the page turns each photograph into a black-and-white outline of the body and marks the shoulders, elbows, wrists, hips and ankles; the photographs themselves never leave the device. The server lines the two outlines up, uses the height to convert pixels into millimetres, and builds a three-dimensional voxel model of the body from them, correcting the common error in which outstretched arms are given the full thickness of the torso. It then slices this model at the chest, waist and hip, measures the length around each slice with clearly stated geometric models, and also derives shoulder length and a hip-to-ankle length. The user receives every measurement together with an explanation of how it was obtained, warnings where a value is less reliable, and an interactive 3D view of the reconstructed body that can be exported as a mesh. The method was checked on synthetic shapes whose true size is known, on a public benchmark of real body silhouettes and on volunteers measured with a tape. For people wearing reasonably fitted clothing the estimates were close enough to tape measurements to support practical uses such as choosing a garment size when shopping online. Loose clothing was found to be the main limitation: because the outline follows the garment rather than the body, a loose shirt makes the chest and waist appear larger, so close-fitting clothing is part of the capture instructions.

Keywords: Anthropometry, body measurement, computer vision, data minimisation, ellipse fitting, privacy preservation, silhouette, visual hull, 3D reconstruction.

DECLARATION

We, hereby declare that the Major Project Phase-2 work entitled "Anthropometric - An Algo Approach" is a bonafide work and has been carried out by us under the guidance of Dr. Jyothi S. Nayak, Professor, Department of Computer Science and Engineering, B.M.S. College of Engineering, Bengaluru, in partial fulfillment of the requirements of the degree of Bachelor of Engineering in Computer Science and Engineering of Visvesvaraya Technological University, Belagavi. We further declare that, to the best of our knowledge and belief, this project has not been submitted either in part or in full to any other university for the award of any degree.

Candidate details:

SL. NO.	Student Name	USN	Student's Signature
1	Sanath S. Shetty	1BM23CS297	
2	Suhas M S	1BM23CS346	
3	Sumanth S. Shetty	1BM23CS348	
4	Thilak K	1WA23CS021	

Place: Bengaluru
Date:

Certified that these candidates are students of Computer Science and Engineering Department of B.M.S. College of Engineering. They have carried out the project work titled "Anthropometric - An Algo Approach" as Major Project Phase-2 work. It is in partial fulfillment for completing the requirement for the award of B.E. degree by VTU. The work is original and duly certify the same.

Guide Name: Dr. Jyothi S. Nayak
Date:
Signature

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Contents

1 Introduction 1
1.1 Overview . . . 1
1.2 Motivation . . . 2
1.3 Objective . . . 3
1.4 Scope . . . 3
1.5 SDG Justification . . . 4
1.6 Existing System . . . 5
1.7 Proposed System . . . 5
1.8 Work Plan . . . 6

2 Literature Survey 8
2.1 Overview . . . 8
2.2 Detailed Survey . . . 8
2.2.1 Manual, scanner and photographic anthropometry . . . 8
2.2.2 Silhouette and visual-hull reconstruction . . . 8
2.2.3 Learning-based body shape and measurement estimation . 9
2.2.4 Clothing as a confounder . . . 9
2.2.5 Privacy-preserving and on-device vision . . . 9
2.3 Research Gap Identification . . . 11
2.4 Mapping of Objectives with Research Gap Identification . . . 11

3 Software and Hardware Requirement Specification 13
3.1 Functional Requirements . . . 13
3.2 Non-functional Requirements . . . 14
3.3 Hardware Requirements . . . 14
3.4 Software Requirements . . . 15
3.5 Cost Estimation . . . 16

4 Designs 18
4.1 System Architecture Overview . . . 18
4.2 Privacy-Preserving System Architecture . . . 19
4.2.1 Threat model: photographs in upload-based processing . . 20

4.2.2 Placement of the privacy boundary 21
4.2.3 Browser-side perception and transmitted data 21
4.2.4 Server-side input validation 22
4.2.5 Data minimisation and enforcement 22
4.2.6 Scope of the privacy property and residual risk 23
4.3 Module/Component Design 23
4.4 Detailed Design 25
4.4.1 Request lifecycle 25
4.4.2 Job states 25
4.4.3 Interface design 26
4.4.4 Processing pipeline design 26
4.4.5 Error handling 27
4.5 Data Design 27
4.5.1 Transmitted data 27
4.5.2 Internal data structures 28
4.5.3 Measurement record 28
4.5.4 Job record and storage 29

5 Methodology 30
5.1 Tools and Techniques 30
5.2 Data Set With Explanation 30
5.3 Flow Of Methodology From Start To End 32
5.4 Explanation Of Methodology In Detail With Equations And Formulae 33
5.4.1 Data acquisition and on-device preprocessing 33
5.4.2 Input layer 33
5.4.3 Weighted pose alignment 34
5.4.4 Metric calibration 34
5.4.5 Conventional visual hull 35
5.4.6 Profile filtering 36
5.4.7 Profile-guided columnwise depth constraint 36
5.4.8 Measurement levels and waist-curvature hypothesis 37
5.4.9 Section extraction and conic recovery 38
5.4.10 Perimeter computation and estimator dependence 39
5.4.11 Shoulder and inseam models 40
5.4.12 Measurement records and display geometry 40
5.4.13 Hyperparameter and constant settings 41
5.4.14 Baseline models 41
5.4.15 Performance measures 43

6 Results and Discussion 45
6.1 Section Models on Synthetic Geometry 45
6.2 Profile-Guided Reconstruction 46
6.3 External Benchmark: BodyM 47
6.4 Volunteer Evaluation with Tape References 48
6.5 Positive and Negative Examples: Effect of Clothing 50
6.5.1 Positive example: subject S8 in fitted clothing 50
6.5.2 Negative example: subject S10 in loose clothing 51
6.5.3 Lessons from the two examples 53
6.6 Privacy Evaluation 53
6.7 Computational Cost 53
6.8 Limitations 54

7 Conclusion and Future Work 55
7.1 Conclusion 55
7.2 Future Work 55

REFERENCES 56
Appendix A: Snapshots 60
Appendix B: Details of list of publications related to this project 64
Appendix C: Details of Patent 65
Appendix D: Details of Funding 66
Appendix E: POs and PSOs Mapped 67
Appendix F: Similarity Report, AI Generated Report 69

List of Figures

1.1 Gantt chart of the project work plan 7

4.1 System architecture 19
4.2 Where the photographs are processed 20
4.3 Job state machine 26
4.4 Framework of the proposed methodology 27

5.1 Metric coordinate frame on a development silhouette pair 35
5.2 Profile-guided reconstruction on a development subject 37
5.3 Waist-curvature hypothesis on a synthetic half-width trace 38
5.4 Section-model comparison on synthetic convex sections 40

6.1 BodyM Test-A results 47
6.2 Signed errors of the volunteer evaluation 50
6.3 Negative example: subject S10 in loose clothing 52

A.1 Front and Side View Image Input 60
A.2 Calculated Measurements 61
A.3 Generated Visual Hull 61
A.4 Extracted Silhouettes from Images 62
A.5 3D Model (STL) Output 63
F.1 Similarity Detection 69
F.2 AI Writing Detection 69

List of Tables

1.1 Routes to body measurement. 5
1.2 Project work plan. 7

2.1 Summary of existing body-measurement and body-shape estimation studies. 10
2.2 Mapping of the identified research gaps to the project objectives. 12

3.1 Hardware requirements. 15
3.2 Software requirements. 16
3.3 Cost estimation. 17

4.1 Modules of the system. 24
4.2 HTTP endpoints of the web service. 26
4.3 Format of the request sent by the client. 27
4.4 Data handled by the web deployment. Items not sent to the server remain in the browser. 28
4.5 Fields of a measurement record. 29

5.1 Tools and techniques used in the project. 30
5.2 Datasets used in the project. 31
5.3 Source geometry and defined outputs. 41
5.4 Hyperparameter and constant settings of the proposed pipeline. . 42
5.5 Comparison analysis of reconstruction and section models. 43

6.1 Test-family MAE of section models against continuous truth . . . 45
6.2 Synthetic occupancy IoU against prescribed truth 46
6.3 BodyM Test-A prediction errors 47
6.4 Volunteer results 49
6.5 Volunteer error statistics 49
6.6 Positive and negative examples compared 50

E.1 Programme Outcomes Mapping with Project 67
E.2 Programme Specific Outcomes Mapping with Project 68

Chapter 1 Introduction

1.1 Overview

Human body measurements underpin garment sizing, ergonomics and clinical screening. Clothing standards define circumferences, breadths and surface paths in terms of anatomical landmarks [1], so that a chest or waist girth means the same thing to a tailor, a manufacturer and a sizing chart, and girths such as waist circumference are also routinely used as indicators of health risk [2]. How these numbers are obtained has changed considerably over time, and each stage of that evolution solved one problem while introducing another.

The oldest and still the reference procedure is manual tape measurement. A trained measurer locates each landmark on the body and wraps a tape around it, which gives a direct reading but requires physical contact, time and skill. Its repeatability depends on how consistently the landmark is identified, how tightly the tape is pulled and how the subject stands, so two measurers, or the same measurer on two occasions, can obtain noticeably different readings [3]. Tape measurement is also inherently in person: it does not scale to large populations and is impractical for someone shopping from home.

Three-dimensional body scanners were introduced to remove the tape and its operator. Laser, structured-light and multi-camera systems capture the whole body surface in seconds, and standardised protocols define how dimensions are extracted from the resulting scans and validated against traditional measurement [4, 5]. Scanning made large anthropometric surveys and automatic size extraction possible, but it replaced one constraint with another: the subject must visit a dedicated booth with specialised, costly hardware and a controlled environment.

Photography was the natural next step, because an ordinary camera, and later a smartphone, is available to almost everyone. Early photographic systems measured breadths and depths from calibrated front and side images and converted them to girths with regression or ellipse models (Section 1.2). With the arrival of deep learning, this idea was extended to networks that predict a complete 3D body, or the measurements directly, from one or two pictures. These methods brought measurement into the home, but they depend on large training sets and usually send full-body photographs to a remote server for processing, which raises its own concerns about where those images go.

The present work continues this evolution. It keeps the convenience of the smartphone photograph and the transparency of classical geometry, but processes

the photographs on the user's own device and computes the measurements without any model trained on measurement labels. Neither tape nor scanner is convenient for a person who wants their measurements at home, for example before buying clothes online, and this project is aimed precisely at that situation.

This project develops a pipeline that needs no task-specific training and no special hardware, and in which the photographs are processed on the user's device. From one frontal and one profile photograph and the stature, it estimates chest, waist and hip girths, shoulder length and a hip-to-ankle length. Pretrained pose and segmentation models [6, 7] run in the browser through WebAssembly [8]; only binary silhouettes, ten landmarks per view and the stature are transferred to the server. The server builds an occupancy volume by silhouette intersection [9] and evaluates it with predefined geometric models, so that every number can be linked to a reconstruction branch, a body position and a section model.

1.2 Motivation

Smartphone photographs are an attractive alternative to tape and scanner because the capture device is widely available. Early photographic systems estimated breadths and depths from calibrated front and side images and converted them to girths through regression or ellipse models [10–13]. More recent work regresses a parametric body model [14] or the measurements themselves from images and silhouettes with deep networks [15–19]. These learning-based methods need large annotated or synthetic training sets and, in a typical web deployment, the upload of full-body photographs to a server.

That upload is the problem this work addresses. A full-body photograph taken for measurement, ideally in close-fitting clothing, shows the subject's face, body shape, skin and home, and the image file may carry metadata such as time and location. Once transmitted, its storage, reuse and deletion are controlled by the operator. Data-protection law in the European Union and India requires that personal data be limited to what is necessary for the purpose [20, 21], and the privacy-by-design principle asks that this be achieved through system architecture [22]. Geometric measurement needs the outline of the body, a few landmarks and the stature, not the appearance of the person.

A second motivation is transparency. Because two silhouettes do not determine a three-dimensional body uniquely, every girth estimated from them rests on choices that the photographs cannot settle: the model used to fill in the depth hidden from both cameras, the body height at which the cross-section is cut, and the shape assumed for that cross-section. This project therefore makes these assumptions explicit, reports competing section models side by side and attaches a

provenance record to every value.

1.3 Objective

The objectives of this project are:

- Estimate chest, waist and hip girths, shoulder length and a hip-to-ankle length from one frontal and one profile photograph and the stature, without training any model on measurement labels.
- Reduce the depth over-estimation that limbs cause in a conventional two-view visual hull.
- Keep the photographs on the user's device and minimise the data transmitted to the server.
- Make every reported value auditable by recording its reconstruction source, level, section model and availability.
- Evaluate the pipeline on controlled synthetic geometry, a public benchmark and volunteers with tape references, and characterise its failure modes, in particular loose clothing.

1.4 Scope

The project is concerned with tailoring and garment sizing: obtaining, without a tape and without a trained measurer, the body measurements that a tailor or a sizing chart asks for. Clothing standards express garment sizes in terms of a small set of body dimensions taken at defined anatomical locations [1]. The system therefore targets the dimensions that matter most for fitting upper- and lower-body garments: the chest, waist and hip girths that decide the size of shirts, jackets, dresses and trousers; the shoulder length that governs the fit of a jacket or shirt across the back; and a hip-to-ankle length that serves as a guide for trouser length. Health and clinical uses of these measurements are not addressed.

The intended user is an individual at home, for example a person buying clothes online or sending measurements to a tailor, who has only a smartphone and a web browser. The user supplies two ordinary photographs, one from the front and one from the side, taken while standing upright with the arms held away from the body, together with their own height. No calibration object, special background or depth sensor is required; the camera only needs to frame the whole body and stay at the same distance for both views.

Within this setting the scope is defined by three principles. First, the photographs are processed on the user's device: only a black-and-white outline of the

body in each view, a few body landmarks and the height are sent to the server, so the appearance of the person is never transmitted. Second, the measurements are obtained by geometry rather than learned from examples: no part of the system is trained on measurement labels, and every stage after the pretrained perception model follows fixed, stated rules. Third, the results are transparent: each value is reported together with how it was produced, and a value that cannot be computed reliably is reported as unavailable with a reason instead of being replaced by a guess.

The scope is also bounded by a theoretical limitation that the project accepts openly. Two outlines record only the width of the body seen from the front and its depth seen from the side; they do not determine the full three-dimensional shape of each cross-section. Any girth computed from them therefore rests on an assumption about the shape that lies between the two views, for example that a cross-section is approximately elliptical. The project makes these assumptions explicit, reports competing shape models side by side, and evaluates how much the answer depends on them, rather than presenting a single number as if the photographs had determined it.

The following lie outside the scope of the present work:

- Camera intrinsics, yaw and capture tolerances are not calculated, and calibrated confidence intervals are not reported.
- The chest, waist and hip measurements are determined using proportional reference levels rather than identifying the actual widest or narrowest points of the body.
- The volunteer evaluation is initial; agreement analysis with matched-plane tape readings on a larger, separate cohort is left for future work.
- The privacy property protects the photographs but not the derived silhouettes, whose retention policy remains to be implemented.

1.5 SDG Justification

The project is aligned with the following United Nations Sustainable Development Goals:

- SDG 9 – Industry, Innovation and Infrastructure. The system replaces a dedicated body scanner with ordinary photographs and openly available perception models executed on the user's own CPU. It requires no training data, no GPU and no specialised hardware, which lowers the cost of adopting body-measurement technology.

- SDG 12 – Responsible Consumption and Production. Clothing sizes are defined in terms of body measurements [1]. Better-informed size choices when buying garments online could reduce returns of wrongly sized clothing and the associated transport and waste; this benefit is a motivation and has not been measured in this project.

1.6 Existing System

Table 1.1 contrasts the existing routes to body measurement. Manual tape measurement requires physical contact and a trained measurer [3], and three-dimensional body scanners rely on dedicated booths [4, 5]. Learning-based photographic systems can be used at home, but they need large training sets and, in a typical web deployment, upload the full-body photographs to a server, where segmentation, pose estimation and measurement are run.

Table 1.1: Routes to body measurement.

Route	Self-use at home	Task-specific training	Raw images uploaded	Main burden
Manual tape	No	No	No	Trained measurer, contact
3D body scanner	No	No	No	Dedicated booth, cost
Cloud deep learning	Yes	Yes	Yes	Training data, photo upload
Proposed	Yes	No	No	Sensitive to loose clothing

In such an upload-based design the photographs reveal far more than the measurements requested from them, and once transmitted their retention and deletion are decided by the operator's systems and policy (Section 4.2.1). On the measurement side, classical photographic methods assume calibrated capture and report a single section model, while silhouette-intersection methods suffer from a weakness of the two-view visual hull: because the profile view cannot separate an arm from the torso behind it, an arm held out to the side is given the full depth of the torso (Section 5.4.7).

1.7 Proposed System

The proposed approach performs body measurements using geometric models while processing the images directly on the user's device (Fig. 4.4). The funda-

mental components (visual hulls, Procrustes alignment, Savitzky–Golay filtering, ellipse fitting and elliptic integrals) are established; the significance lies in how they are integrated and assessed:

- A profile-guided columnwise depth constraint: a filtered profile interval provides a sagittal midpoint and depth, and the vertical run length of each frontal silhouette column restricts local depth. This limits the replicated torso depth that a horizontal limb acquires in the conventional hull, without a learned prior. It is a geometric constraint, not a unique reconstruction of the body.
- Conditional geometric measurement with provenance: alternative section models are reported separately, with an exact elliptic-integral perimeter and clearly stated unavailability, and a deletion-only waist-curvature hypothesis is kept as a separate record.
- Data minimisation by architecture: perception runs on the device and the server permits only strictly binary silhouettes, landmarks and stature (Section 4.2, Fig. 4.2).
- A three-tier evaluation (synthetic geometry, BodyM benchmark [18] and volunteer tape references), including documented positive and negative examples showing how fitted clothing supports accurate girths and loose clothing inflates them.

1.8 Work Plan

The project was organised into the seven phases of Table 1.2, each ending with a reviewed deliverable.

Table 1.2: Project work plan.

Phase	Work carried out	Period
1	Review of photographic, scanner and learning-based anthropometry; problem formulation; choice of measurement sites	Feb – Mar 2026
2	Silhouette and landmark extraction, weighted Procrustes alignment and stature-based metric calibration	Mar – Apr 2026
3	Conventional visual hull and the profile-guided columnwise depth constraint	May – Jul 2026
4	Section extraction, ellipse fitting, elliptic-integral perimeters, waist-curvature hypothesis, shoulder and inseam models	Jul – mid Aug 2026
5	Web application: browser-side perception, server-side input validation, compute queue and 3D viewer	mid Aug – Sep 2026
6	Synthetic experiments, BodyM benchmark and volunteer evaluation with tape references	Sep – Oct 2026
7	Research manuscript and project report	Oct – Nov 2026

Fig. 1.1 shows the same plan as a Gantt chart. Each bar spans the period of Table 1.2 at half-month resolution, so overlapping phases, such as the start of the web application while the measurement models were being completed, are visible directly.

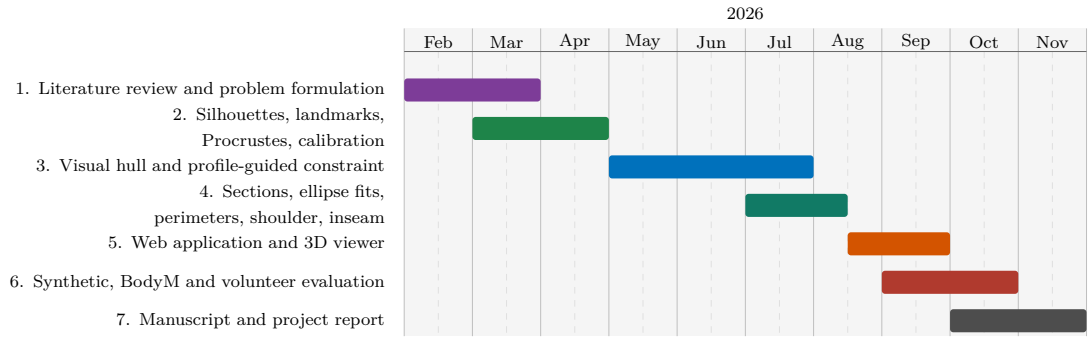


Figure 1.1: Gantt chart of the project work plan (Table 1.2). Columns are months of 2026, subdivided into halves by dashed lines.

Chapter 2 Literature Survey

2.1 Overview

Estimating body dimensions without a tape has been studied from three directions: photographic and silhouette geometry, learned body-shape regression and, more recently, privacy-aware on-device vision. The survey below also covers the standards that define body measurements and the confounding effect of clothing. Table 2.1 summarises representative studies.

2.2 Detailed Survey

2.2.1 Manual, scanner and photographic anthropometry

ISO 8559-1 defines the landmarks and paths of clothing measurements [1], and ISO 20685-1 specifies how dimensions extracted from 3D scans should be validated against traditional measurement [4]. Ulijaszek and Kerr reviewed the sources and magnitude of manual measurement error [3]. Bartol et al. surveyed body measurement from 3D scanning and noted the cost and controlled environment of scanners [5], Heymsfield et al. reviewed digital anthropometry for clinical use [2], and Yan et al. extracted clothing measurements automatically from complete 3D scans [23]. Choudhary et al. validated a smartphone application that estimates the waist-to-hip ratio from front, side and back photographs against trained tape measurement in 1200 participants, with an error of about 0.015 in the ratio [24].

With photographs, Meunier and Yin evaluated a 2D image-based measurement and sizing system using front and side views [10]. Hung et al. combined photographic breadth and depth with ellipse-based circumference models [11], and Lin and Wang constructed a 3D human model from front and side images [13]. Aslam et al. automated anthropometric measurement from frontal and lateral silhouettes with an ellipsoid section model under calibrated capture [12]. These studies show that two orthogonal views carry much of the information needed for torso girths, but they assume calibrated capture, report a single section model and do not consider where the photographs are processed.

2.2.2 Silhouette and visual-hull reconstruction

Laurentini formalised the visual hull as the largest volume consistent with a set of silhouettes [9]. With only two orthogonal views, the hull at each height pairs every frontal run with every profile run, so a horizontal arm attached to the torso

inherits the full torso depth. Boisvert et al. inferred 3D shape from front and side silhouettes by fitting a statistical shape space learned from scans [25], and recent compact multi-view systems combine RGB and depth imagery with SMPL fitting [26]. The approach proposed here constrains the two-view hull with a geometric rule derived from the two silhouettes themselves.

2.2.3 Learning-based body shape and measurement estimation

Parametric models such as SMPL represent body shape in a low-dimensional statistical space [14]. HMR regresses SMPL parameters directly from a single image [15], and BodyNet infers a volumetric body [27]. Closer to anthropometry, Dibra et al. regressed body shape from silhouettes with CNNs [16], Smith et al. improved silhouette-based reconstruction accuracy [17], Ji et al. learned silhouette features for shape reconstruction [28], and Choutas et al. used metric and semantic attributes to regress body shape [19]. Ruiz et al. presented the BodyM dataset of real silhouettes with scan-derived measurements and trained a measurement regressor with adversarial body simulation [18]. These approaches produce promising outcomes; however, they depend on extensive annotated datasets, and their practical implementation generally involves processing photographs on remote servers.

2.2.4 Clothing as a confounder

All silhouette methods represent the clothed envelope, not the skin. Bălan and Black established that body shape beneath clothing can be estimated by using multiple poses and considering that the body remains within the garment boundaries. [29]. For this reason, two-view photographic techniques, such as the BodyM capture protocol [18], necessitate close-fitting garments, although studies rarely provide quantitative evidence of how loose clothing affects the calculated girths.

2.2.5 Privacy-preserving and on-device vision

Moving inference to the device retains raw images at their source. In the browser, WebAssembly makes near-native on-device inference practical [8], and MediaPipe provides pose and segmentation models designed for on-device use [6, 7, 30]. Data-protection law states the data-minimisation principle [20, 21], and privacy by design asks that it be built into the system [22]. To our knowledge, these elements have not been combined in an anthropometric pipeline whose server interface accepts only binary silhouettes.

Table 2.1: Summary of existing body-measurement and body-shape estimation studies.

Ref	Year	Approach	Input	Task-specific training	On-device privacy design	Limitation
[10]	2000	2D image-based measurement and sizing	Front + side photos	Regression on measured sample	No	Controlled capture; regression-based girths
[11]	2004	Breadth/depth with ellipse girth model	Front + side photos	None	No	Single ellipse assumption; small sample
[13]	2012	3D model from front and side images	Front + side photos	Template model	No	Template-dependent; controlled backdrop
[25]	2013	Statistical shape fitting to silhouettes	Front + side silhouettes	Scan-based shape space	No	Requires a 3D scan database prior
[16]	2016	CNN shape regression (HS-Nets)	1–2 silhouettes	Large synthetic set	No	Synthetic training; minimal clothing assumed
[12]	2017	Fiducials on silhouettes + ellipsoid	Front + lateral silhouettes	None	No	Calibrated capture; one section model
[15]	2018	End-to-end SMPL regression (HMR)	Single RGB image	2D/3D pose datasets	No	Shape and metric scale weakly constrained
[27]	2018	Volumetric body inference (BodyNet)	Single RGB image	Synthetic (SURREAL)	No	Voxel resolution; synthetic-to-real gap
[17]	2019	Silhouette-based body reconstruction	Multiple silhouettes	Synthetic training	No	Learned prior; clothing not modelled
[28]	2019	Learned silhouette features	Binary silhouettes	Body shape dataset	No	Requires labelled shapes
[19]	2022	Shape regression with metric/semantic attributes	Single RGB image	Attribute-labelled data	No	Annotation effort; server-class model
[18]	2022	Measurement regression with adversarial body simulation	Front + side silhouettes, height, weight	Synthetic + BodyM	No	Training data; tight clothing required
[26]	2025	Compact multi-view imaging + SMPL	Multi-view RGB(-D)	Model fitting with priors	No	Dedicated multi-camera rig
Proposed	2026	Profile-guided visual hull + geometric estimators	Front + side photos, stature	None (pretrained perception only)	Yes	Proxy measurement levels; sensitive to loose clothing

2.3 Research Gap Identification

Table 2.1 shows a clear gap. Classical photographic methods are training-free and understandable but assume calibrated capture and a single section model, and they leave the limb-induced depth errors of two-view intersection unaddressed. Learning-based methods reach high accuracy on benchmarks but rely on large, often synthetic, training sets and on server-side inference, so the full-body photographs leave the subject's control. None of the surveyed studies treats the privacy of the raw photographs as a design requirement, none reports competing section models side by side with their availability, and the effect of loose clothing on girth estimates is rarely quantified. There is therefore a need for a measurement pipeline that (i) needs no task-specific training, (ii) constrains two-view reconstruction geometrically, (iii) processes the photographs on the device and (iv) reports each measurement with its provenance and documented failure modes.

2.4 Mapping of Objectives with Research Gap Identification

Table 2.2 links each gap identified in Section 2.3 to the objective that responds to it and to the part of the report where the response is described and evaluated.

Table 2.2: Mapping of the identified research gaps to the project objectives.

Research gap	Objective	Response in this project
Learned estimators depend on large labelled or synthetic training sets	Estimate the measurements without training any model on measurement labels	Pretrained perception only; all measurement stages are geometric and use fixed constants listed in the Methodology chapter
Two-view intersection replicates torso depth into limbs, and the error is left untreated	Reduce the depth over-estimation caused by limbs in the visual hull	Profile-guided columnwise depth constraint, described in the Methodology chapter and evaluated on synthetic shapes and the BodyM benchmark
Full-body photographs are uploaded and leave the subject's control	Keep the photographs on the device and minimise the transmitted data	Browser-side perception and a server that accepts only binary silhouettes, landmarks and stature, described in the Design chapter
Studies report one number per site without stating the reconstruction and section model behind it	Make every reported value auditable	Competing section models reported separately, each value carrying a provenance record
The failure modes, in particular loose clothing, are rarely quantified	Evaluate the pipeline and characterise its failure modes	Synthetic, BodyM and volunteer evaluation, with documented positive and negative clothing examples in the Results chapter

Chapter 3 Software and Hardware Requirement Specification

3.1 Functional Requirements

The functional requirements follow from the scope defined in Section 1.4 and from the behaviour of the two tiers of the system.

1. Input capture. The user chooses a front-facing and a side-facing photograph, taken roughly at right angles, and types in their height in cm. Each photograph has to show the subject from the top of the head down to the feet.

2. On-device perception. The browser decodes each photograph locally and computes a binary silhouette and ten landmarks per view (shoulders, elbows, wrists, hips and ankles). A photograph is rejected on the device if more than three of the ten landmarks have visibility below 0.3.

3. Minimal transmission. The client sends only one binary PNG silhouette per view, ten normalised landmarks per view, the stature and a preprocessing-version tag; photographs, crops, probability maps and file metadata are never sent.

4. Server-side validation. The server accepts a silhouette only if every pixel is 0 or 255, its smaller side is at least 200 pixels and the foreground covers 2–85% of the frame; landmarks must be complete and in $[0, 1]$, and the stature must lie between 50 and 260 cm. A rejected request is reported to the user and its job directory is deleted.

5. Alignment and calibration. The two views are aligned by weighted Procrustes analysis and scaled to millimetres using the stature; excessive rotation or scale causes rejection.

6. Reconstruction. The server builds both the conventional visual hull and the profile-guided reconstruction from the same inputs.

7. Measurement. Chest, waist and hip girths are measured at prescribed levels with the constrained ellipse, the width/depth ellipse and the convex-hull diagnostic; the shoulder path or breadth and a hip-to-ankle proxy are also produced. The optional waist-curvature hypothesis is stored as a separate record.

8. Results and provenance. Each value is returned with its reconstruction branch, level, section model, fallback reason, flags and availability, together with an interactive 3D view of the reconstruction; a surface mesh can be exported on request.

3.2 Non-functional Requirements

- Privacy and data minimisation. The data of the user should not be uploaded to the server , and the server should reject any non-binary image (Section 4.2).
- Transparency and auditability. No measurement value should be replaced by a default value , if values are unavailable report it with reason.
- Reproducibility. All of the constants are fixed and are the same for every subject (Table 5.4).
- Hardware independence. Client side inference must run on even low end devices (laptop ,phone etc) without a dedicated GPU or depth sensors of any kind
- Performance. An end to end run must be completed within 3 minutes
- Robustness. Reasons for rejection of an image due to issues such as missing pose or segmentation, clipping, border contact, fragments, empty input, excessive grid size or invalid stature etc must be clear and explicit rather than crashes or errorenous values
- Security of transport. The server is deployed behind HTTPS.

3.3 Hardware Requirements

Because perception runs on the client CPU, the client needs no special hardware; the server holds dense occupancy grids in memory during reconstruction. Table 3.1 lists the hardware used and recommended.

Table 3.1: Hardware requirements.

Component	Specification
Camera	Any smartphone or digital camera that can frame the whole body; both views taken at the same distance and zoom, ideally from a tripod or a fixed, level support
Client device	Phone, tablet, laptop or desktop with a WebAssembly-capable browser; inference uses the CPU, so a GPU is not required
Development / server machine	AMD Ryzen 7 5000-series processor with 16 GB RAM running Windows; the recorded runs peaked at 2.22 GB and 1.80 GB of working memory, so a minimum of 4 GB RAM is advisable
Storage	Disk space for one directory per run containing the received silhouettes, landmarks, logs, measurements and reconstructions
Network	Internet connectivity, with the server published over HTTPS

3.4 Software Requirements

Python is used for both the research pipeline and the server, and the browser client executes an identical, version-pinned pose model compiled to WebAssembly. The software stack is listed in Table 3.2.

Table 3.2: Software requirements.

Software (version)	Used for
Python 3.12.9	Research pipeline and server-side alignment, reconstruction and measurement
MediaPipe Pose Landmarker (heavy), Python	Silhouettes and landmarks in the Python research path
MediaPipe Tasks Vision (WebAssembly)	In-browser segmentation and landmark detection; runtime loaded from a public CDN, model served by the application
OpenCV	Contour extraction, connected components and morphology
NumPy 2.5.2 and SciPy 1.18.1	Voxel grids, SVD for Procrustes alignment, signal smoothing and the elliptic-integral perimeter
scikit-image 0.26.0 and trimesh 5.1.0	Marching cubes, mesh smoothing and simplification for the display surface
Flask with Gunicorn (Docker)	Web interface, submission and status endpoints
Web browser	Client tier; must support WebAssembly
Operating system	Windows for development and the recorded runs

3.5 Cost Estimation

No component of the system had to be purchased. All libraries and the perception model are open source, the BodyM dataset is free for research under its Creative Commons Attribution–NonCommercial 4.0 licence, development was done on the students' own laptops, and the deployment uses free hosting and a free domain with an HTTPS certificate. Inference runs on the user's own device, so there is no recurring GPU or cloud-inference cost. Table 3.3 summarises the estimate.

Table 3.3: Cost estimation.

Item	Source / licence	Cost (INR)
Python, MediaPipe, OpenCV, NumPy, SciPy and other libraries	Open source	0
BodyM benchmark dataset	CC BY-NC 4.0	0
Development computers	Students' existing laptops	0
Server hosting	syslk.space	0
Domain name and HTTPS certificate	syslk.space	150
Total		150

Chapter 4 Designs

This chapter describes the design of the system before the methodology of Chapter 5 explains the theory and mathematics of each stage. It proceeds from the outside in: the overall client–server architecture; the privacy-preserving architecture that decides where the photographs are processed and what may cross to the server; the modules that make up each tier; the detailed behaviour of a single measurement request, including the end-to-end processing flow; and finally the data that are created, exchanged and stored. All block diagrams of the system are collected in this chapter.

4.1 System Architecture Overview

The system follows a two-tier client–server architecture in which the tier boundary is also the privacy boundary. The client tier is a web page that runs in the user's browser. It owns the photographs and performs every operation that needs their pixels: decoding, segmentation and landmark detection with the MediaPipe Pose Landmarker compiled to WebAssembly. The server tier is a Python web service packaged in a Docker container and served by Gunicorn behind HTTPS. It receives only binary silhouettes, landmarks and the stature, validates them, and runs the geometric pipeline of alignment, reconstruction and measurement. Results are returned to the browser as measurement records together with an interactive 3D viewer.

The server tier is itself organised in three layers. The interface layer exposes the HTTP endpoints and performs input validation; the job layer assigns each request an identifier and a working directory and admits it to a bounded compute queue; and the processing layer executes the research pipeline as a separate process for each job, so that a failure in one run cannot affect the web service or other runs. Fig. 4.1 shows the architecture.

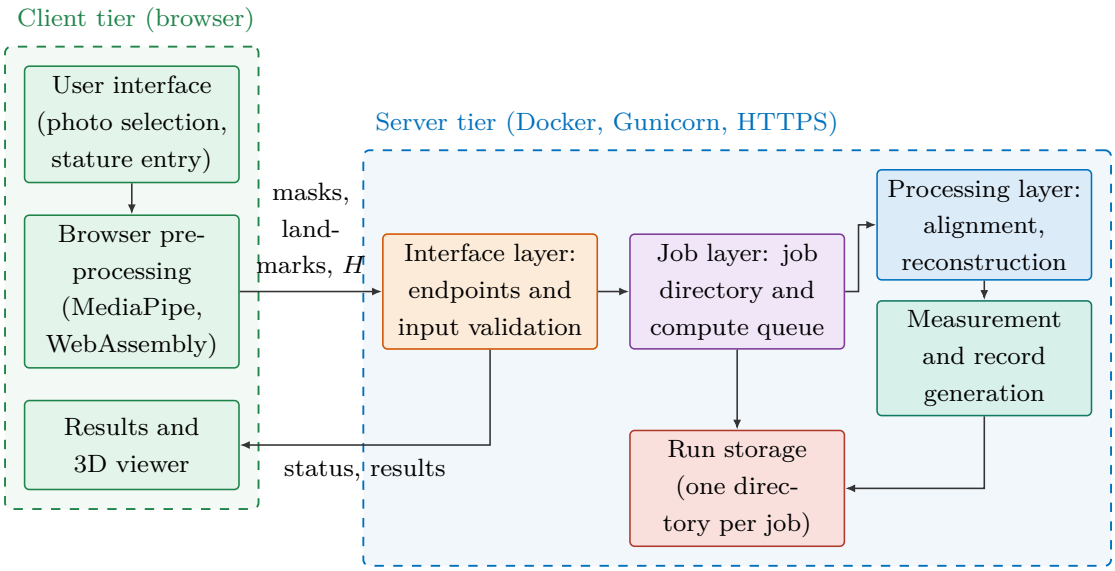


Figure 4.1: System architecture. The client tier keeps the photographs on the device and sends only derived inputs. The server tier is layered into interface, job and processing layers; every run writes to its own directory, from which the results and the 3D viewer are served.

The main design decisions are summarised below.

- Perception on the client. Placing segmentation and landmark detection in the browser keeps the photographs off the server and removes any need for server GPUs.
- Geometry on the server. Reconstruction holds dense occupancy grids in memory, which is better suited to a server than to a phone or a browser tab.
- Asynchronous jobs. A request returns a job identifier immediately and the browser polls for progress, so long runs do not hold open an HTTP connection.
- Process isolation. Each run is a separate operating-system process working inside its own directory, which bounds the memory of each run and keeps the web service responsive.
- Bounded admission. The compute queue admits a fixed number of running and pending jobs and refuses further requests instead of buffering them without limit.

4.2 Privacy-Preserving System Architecture

The architecture of Section 4.1 places the tier boundary where it does for a reason: keeping the photographs on the user's device is one of the project objectives, and it

is treated as a design requirement that is assessed alongside accuracy. The privacy property is obtained from where each operation runs, not from a policy promise. Fig. 4.2 contrasts the conventional upload-based design with the architecture of this system, and Table 4.4 in the data design (Section 4.5) lists every data item, where it is processed and whether it is sent to the server.

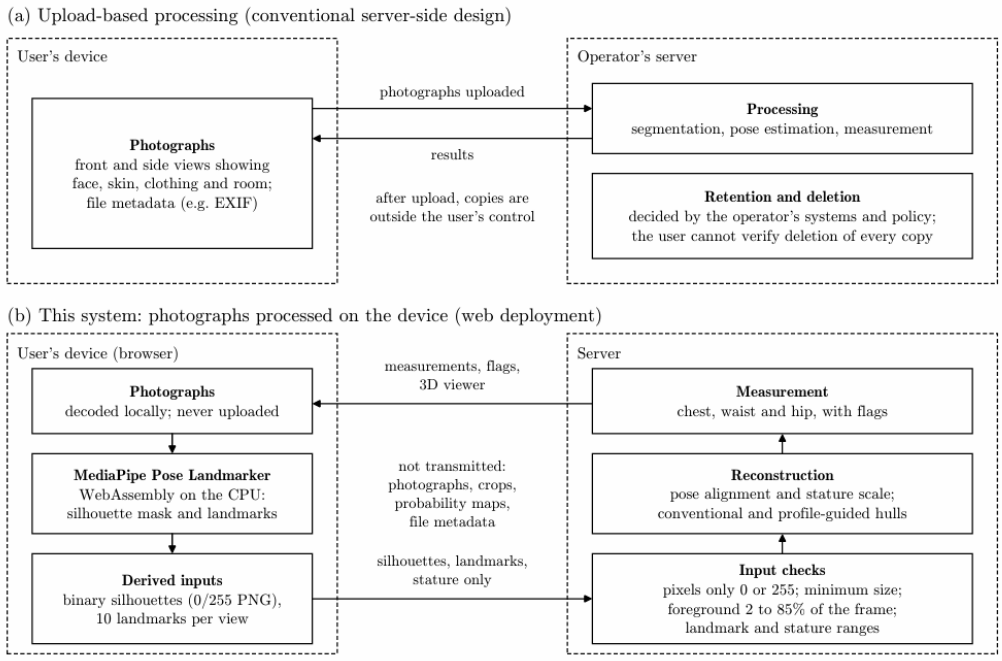


Figure 4.2: Where the photographs are processed. (a) In a conventional upload-based design the photographs themselves reach the operator's server, which runs segmentation, pose estimation and measurement; after upload, their retention and deletion are decided by the operator's systems and policy and are outside the user's control. (b) In the web deployment of this system the photographs are decoded in the browser, and MediaPipe Pose Landmarker runs on the device's CPU through WebAssembly to produce a silhouette mask and landmarks. Only binary silhouettes (0/255 PNG), ten landmarks per view and the stature are sent; photographs, crops, probability maps and file metadata are not transmitted. The server checks the inputs, rejecting any silhouette whose pixels are not strictly 0 or 255, reconstructs and measures the body, and returns measurements, reliability flags and a 3D viewer. The transmitted data still describe body shape (Section 4.2.6).

4.2.1 Threat model: photographs in upload-based processing

A conventional implementation of photographic anthropometry, including the earlier server-side mode of this pipeline, uploads the frontal and profile photographs and runs segmentation, pose estimation and measurement on a server (Fig. 4.2a). The photographs reveal far more than the measurements requested from them. Taken in close-fitting clothing, they show the subject's face, body shape and skin together with the room in which they were taken, and image files can carry meta-

data such as capture time, device model and location. Once a photograph has been transmitted, the user has no technical control over it: whether it is stored, written to logs or backups, viewed by staff, passed to third-party processors, reused or deleted is decided by the operator's systems and policies, and the user cannot independently verify that every copy has been removed. A later breach of the server exposes whatever has been retained. Contractual and regulatory safeguards can limit these risks, but they do not return control of a transmitted image to its subject. Data-protection law expresses the relevant principle as data minimisation: personal data should be "adequate, relevant and limited to what is necessary" for the purpose of processing [20,21], and privacy by design asks that this be achieved through the architecture of the system [22]. Geometric measurement needs the outline of the body, a few landmarks and the stature, not the appearance of the person. The adversary considered here is therefore any party with access to the server side (the operator, its staff and processors, or an intruder after a breach), and the goal is that such a party never obtains the photographs in the normal workflow.

4.2.2 Placement of the privacy boundary

The privacy boundary coincides with the tier boundary of Fig. 4.1 (see also Fig. 4.2b) and is placed immediately after perception, the last stage that needs image pixels. Throughout the report, I_f and I_s denote the frontal and profile photographs, M_f and M_s the corresponding binary silhouettes, Λ_f and Λ_s the two sets of ten landmarks, and H the stature. Everything that needs I_f or I_s happens on the client; everything on the server works only from M_f, M_s, Λ_f, Λ_s and H.

On the client, the web page loads the MediaPipe Tasks Vision WebAssembly runtime [8, 30] and the same pinned heavy Pose Landmarker model that is used by the Python research pipeline [6, 7]; the model file is served by the application server itself, so the model version is fixed. Inference runs on the CPU and needs no GPU. On the server, the interface layer validates the request (Section 4.2.4) before any job is created, and the processing layer runs the geometric pipeline of Fig. 4.4. No stage of the server tier has access to the photographs, so none of them can store, log or forward them.

4.2.3 Browser-side perception and transmitted data

The perception steps of the client tier (resizing, segmentation, mask clean-up and landmark refinement) are specified in Section 5.4.1. From a design point of view, two of their properties matter. First, if the cropped landmark refinement fails, the landmarks from the full image are kept, so a refinement failure never blocks a

request. Second, a photograph is rejected on the device if more than three of the ten required landmarks are poorly visible, so an unusable capture is caught before anything is sent.

The request contains only the items marked as transmitted in Table 4.4: one binary PNG silhouette per view, ten normalised landmarks per view, the stature and a preprocessing-version tag. Each silhouette is encoded from the cleaned binary array rather than derived from the source file, so it carries no pixel values and no metadata of the photograph. The subject crops, probability maps and decoded photographs exist only in browser memory.

4.2.4 Server-side input validation

Before any computation the server decodes each silhouette M_v of size $h \times w$ as a single-channel image and accepts it only if

$$\begin{aligned} M_v(x,y) \in \{0,255\} \quad \forall (x,y), \\ \min(h,w) \geq 200, \qquad 0.02 \leq \phi_v \leq 0.85, \end{aligned} \tag{4.1}$$

where ϕ_v is the fraction of foreground pixels: every pixel must be 0 or 255, the smaller image dimension must be at least 200 pixels, and the foreground must occupy between 2% and 85% of the frame. The ten landmarks of each view must be present with coordinates and visibilities in $[0,1]$, the stature must lie between 50 and 260 cm, and the total request size is bounded. If any check fails, the request is rejected with a message to the user and its job directory is deleted, so no partial input is retained. A greyscale or colour photograph submitted through this interface by mistake therefore fails the first condition of Eq. (4.1) instead of being silently thresholded and processed. Alignment, metric scaling, reconstruction and section measurement then operate only on these derived inputs.

4.2.5 Data minimisation and enforcement

A binary silhouette stores one bit of shape per pixel, against 24 bits of colour in the photograph, and contains no face, skin tone, clothing colour, texture or background; the landmarks are joint positions, not appearance. Each request carries only M_f, M_s, Λ_f, Λ_s and H, never I_f or I_s. We quantify the reduction by the payload ratio

$$r = \frac{|I_f| + |I_s|}{|M_f| + |M_s| + |\Lambda_f| + |\Lambda_s| + |H|}, \tag{4.2}$$

where $|\cdot|$ is the encoded size in bytes; it is evaluated in Section 6.6. Size is nonetheless the weaker argument: what matters is that the transmitted content

excludes every attribute of the photograph that the measurement does not need.

The binary-value check prevents accidental photograph upload, but it is not client attestation: the server cannot verify that the client executed the published page, and a modified client could submit an arbitrary binary mask. Regression tests verify that the browser's mask clean-up, probability resampling and crop mapping match the Python reference implementation and that, during the data-transfer step, the browser uploads only binary silhouettes and landmarks. The experiments of Chapter 6 use a separate, frozen research implementation with either the Python perception path or supplied silhouettes; they characterise the evaluated research pipeline, while the browser-specific processing is covered by these regression tests.

4.2.6 Scope of the privacy property and residual risk

The design reduces the data that leave the device; it does not make the remaining data non-personal. Silhouettes, landmarks and stature describe body shape and size and are therefore treated as personal, body-derived data, and the server derives further body-shape data from them: occupancy volumes, the measurements, an interactive viewer and an optional surface mesh. In the prototype each run's inputs and outputs are stored in a server directory named by its random identifier, without user accounts and without automatic expiry, so the retention and deletion of these derived data remain under the operator's control. Transport encryption relies on deployment behind HTTPS. The page fetches the WebAssembly runtime from a public content-delivery network, which observes ordinary request metadata but receives no image content. The architecture therefore keeps the photographs off the server in the normal workflow and minimises what is transmitted; it does not guarantee complete privacy, and the derived data must still be protected by the operator.

4.3 Module/Component Design

The system is decomposed into modules with a single responsibility each. Table 4.1 lists them with their tier, responsibility and main inputs and outputs.

Table 4.1: Modules of the system.

Module	Tier	Responsibility	Input → output
Web page	Client	Photo selection, stature entry, progress display, results and 3D viewer	User input → request
Browser preprocessing	Client	Decoding, resizing, segmentation, mask clean-up, landmark refinement	Photographs → masks, landmarks
Web service	Server	Endpoints, input validation, job records, launching runs, serving results	Request → job identifier, status
Compute queue	Server	Bounded admission and execution of heavy jobs	Job → accepted / refused
Pipeline driver	Server	Orchestrates the stages of one run and writes the measurement records	Job inputs → measurements.json
Pose normalisation	Server	Canonical targets and weighted Procrustes alignment	Mask, landmarks → aligned mask
Visual hull	Server	Metric scaling, conventional hull O_0, slicing	Aligned masks, H → O_0
Profile carving	Server	Profile filtering and profile-guided reconstruction O_p	Aligned masks, H → O_p
Waist hypothesis	Server	Optional waist-curvature record	Frontal trace → separate record
Measurement	Server	Levels, sections, convex hull, ellipse fits, perimeters, shoulder and inseam	O_0, O_p → values with provenance
Viewer and mesh export	Server	Interactive 3D viewer and optional surface mesh (display only)	Occupancy → viewer, STL

The modules interact through narrow interfaces. The client communicates with the server only through the HTTP endpoints of Table 4.2. The web service communicates with the pipeline only through command-line arguments and the files in the job directory, and the pipeline modules communicate through NumPy arrays held in memory. No server module imports a perception module in the web deployment, so there is no code path on the server by which a photograph could be processed.

4.4 Detailed Design

4.4.1 Request lifecycle

A single measurement request passes through the following steps.

1. The user selects the frontal and profile photographs and enters the stature. The page decodes each photograph in browser memory.

2. For each view, the browser computes the binary silhouette and ten landmarks as described in Section 5.4.1. A view with more than three landmarks below visibility 0.3 is rejected on the device and nothing is sent.

3. The page sends one multipart POST request to /api/run containing the two silhouettes, the two landmark sets, the stature and the preprocessing-version tag.

4. The web service checks the stature, creates a job directory named by a random 12-character identifier and validates each silhouette and landmark set. On failure the directory is deleted and an error message is returned.

5. The job is submitted to the compute queue. If the queue is full the directory is deleted and the request is refused with HTTP status 429; otherwise the job identifier is returned to the browser.

6. A worker thread launches the pipeline driver as a child process in the job directory. Its output is parsed into progress steps, log lines and reliability flags.

7. On completion the measurement records are loaded, the interactive viewer is exported and the job state becomes done.

8. The browser polls /api/job/<id> until the job is finished and then displays the measurements, flags and 3D viewer. A surface mesh can be requested separately.

4.4.2 Job states

Each job moves through a small state machine (Fig. 4.3). A job is queued when admitted, running once a worker starts it, and ends in either done or error. Errors from the pipeline, a missing results file or an unexpected exception all lead to the error state with a message, so a failed run is always reported rather than left pending.

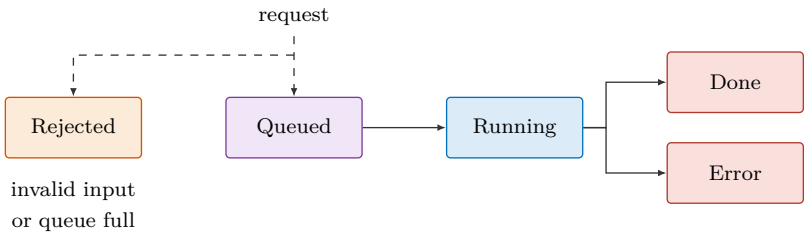


Figure 4.3: Job state machine. A request that fails validation or finds the queue full is rejected and leaves no job directory; an admitted job always ends in done or error.

4.4.3 Interface design

The HTTP interface is kept small. Table 4.2 lists the endpoints exposed by the web service.

Table 4.2: HTTP endpoints of the web service.

Endpoint	Method	Purpose
/	GET	Serves the web page
/models/pose_landmarker_heavy.task	GET	Serves the pinned pose model to the browser
/api/run	POST	Accepts masks, landmarks and stature; returns a job identifier
/api/job/<id>	GET	Returns job state, progress, flags and results
/api/job/<id>/stl	GET, POST	Reports or starts the optional surface-mesh export
/runs/<id>/<file>	GET	Serves the viewer and output files of a run

4.4.4 Processing pipeline design

Inside a job, the pipeline driver calls the server modules in a fixed order: pose normalisation of each view, metric calibration, construction of the conventional hull O_0 and the profile-guided reconstruction O_p, location of the measurement levels, section extraction and the three section models, the shoulder and inseam models, and finally the optional waist hypothesis. Each stage either returns its result or records an explicit reason for unavailability, which is carried into the measurement record. Reconstruction is computed in chunks of 64 columns to bound peak memory, and the display mesh is generated by a separate branch that never feeds back into measurement. Fig. 4.4 shows this end-to-end processing flow across the two tiers; the theory behind each stage is developed in Chapter 5.

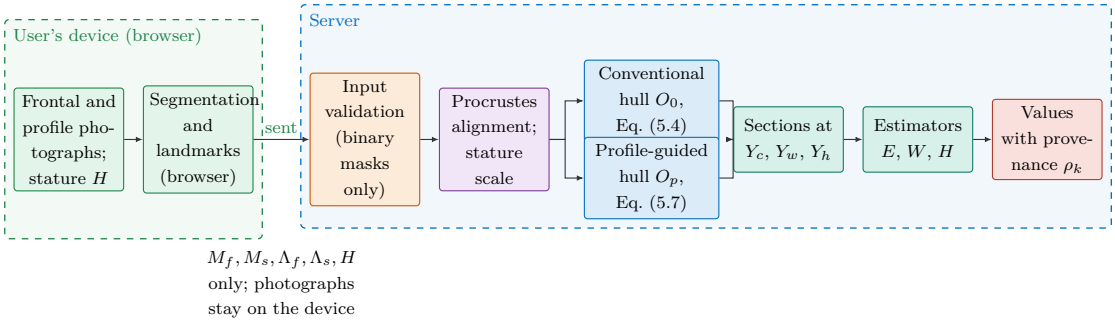


Figure 4.4: Framework of the proposed methodology. Green stages run in the user's browser; the server (blue zone) performs input validation (orange), alignment and metric scaling (purple), reconstruction by two branches (blue), section measurement (teal) and output with provenance (red). Only binary silhouettes, landmarks and stature cross from the device zone to the server zone. Both reconstruction branches are measured at every requested level and reported separately. The client–server architecture is detailed in Fig. 4.2 and Section 4.2.

4.4.5 Error handling

Errors are handled at the earliest stage that can detect them. Unusable photographs are rejected on the device; malformed or non-binary inputs and invalid statures are rejected by the interface layer; excessive load is refused by the compute queue; and geometric failures such as excessive Procrustes rotation, border contact or empty sections are recorded by the pipeline as unavailable values with a reason. No stage substitutes a default value for a failed measurement.

4.5 Data Design

4.5.1 Transmitted data

The request from the client contains only the items marked as transmitted in Table 4.4. Table 4.3 gives their format.

Table 4.3: Format of the request sent by the client.

Field	Type	Constraint
front_mask, side_mask	PNG image	Single channel, values 0 or 255, smaller side ≥ 200 px, foreground 2–85%
front_landmarks, side_landmarks	JSON list	Ten points of $(x, y,$ visibility$)$, all in $[0, 1]$
height	Number (cm)	$50 \leq H \leq 260$
preprocessing	Text tag	Version of the browser preprocessing

Table 4.4: Data handled by the web deployment. Items not sent to the server remain in the browser.

Data item	Processed in	Sent to server
Frontal and profile photographs	Browser	No
Subject crops for landmark refinement	Browser	No
Segmentation probability maps	Browser	No
Source-file metadata (e.g. EXIF)	Not used	No
Binary silhouette per view (PNG, 0/255)	Browser	Yes
Ten landmarks per view (x, y, visibility)	Browser	Yes
Stature; preprocessing-version tag	User entry	Yes
Occupancy volumes, measurements, flags, viewer, optional mesh	Server	No (generated there)

4.5.2 Internal data structures

The aligned silhouettes are held as two-dimensional binary arrays, and the landmarks as arrays of normalised coordinates with visibility weights. Each reconstruction is a three-dimensional boolean occupancy grid over (X, Y, Z) in millimetres, stored with its three coordinate axes and voxel spacing h. A cross-section is a two-dimensional boolean slice of this grid at a measurement level, from which the contour, convex hull and fitted ellipse parameters (centre, semi-axes and orientation) are derived.

4.5.3 Measurement record

Every requested measurement produces one record, whether or not a value could be computed. Table 4.5 lists its fields.

Table 4.5: Fields of a measurement record.

Field	Meaning
Attempt identifier	Identifies the run that produced the record
Site and value	Measurement name (e.g. chest, waist, hip) and value, or null when unavailable
Units	Unit of the value (cm)
Source branch	Reconstruction used: O_0 or O_p
Level	Height Y of the section from the crown
Section model	Estimator used: E, W or H, or the shoulder/inseam method
Fallback reason	Why a fallback method or null value was used
Flags	Reliability flags raised during the run
Availability	Whether the value could be computed

4.5.4 Job record and storage

For each job the web service keeps an in-memory record containing the identifier, state, progress, current step, user-visible log, error message, results, flags, stature, a viewer-availability marker and the creation time; this record is what the status endpoint returns. On disk, each run has its own directory named by the job identifier, holding the received silhouettes (front_mask.png, side_mask.png), the landmark files (front_landmarks.json, side_landmarks.json), the run log, the measurement records (measurements.json), the interactive viewer and, when requested, the surface mesh. There are no user accounts and no database; the directory is the unit of storage, and a rejected request leaves no directory behind. The privacy implications of this storage are discussed in Section 4.2.6.

Chapter 5 Methodology

5.1 Tools and Techniques

Table 5.1 lists the main tools and techniques and the stage in which each is used. All of them are established methods; the contribution of the project lies in how they are coupled and evaluated.

Table 5.1: Tools and techniques used in the project.

Tool / technique	Purpose in the pipeline	Ref.
MediaPipe Pose Landmarker (WebAssembly and Python)	Segmentation mask and ten body landmarks per view	[6, 7, 30]
WebAssembly	Near-native inference inside the browser	[8]
Weighted orthogonal Procrustes analysis	Similarity alignment of each view to canonical landmark targets	[31]
Visual hull	Conventional two-view occupancy O_0	[9]
Savitzky–Golay filter	Smoothing of the profile depth endpoints	[32]
Monotone-chain convex hull	Convex hull and perimeter diagnostic of each section	–
Halíř–Flusser ellipse fit	Ellipse-specific least-squares fit of each section	[33, 34]
Complete elliptic integrals	Exact ellipse perimeter and its sensitivity	[35]
Marching cubes	Display surface for the 3D viewer (not measured)	[36]

5.2 Data Set With Explanation

The pipeline has no training stage, so the data serve only for development and evaluation. Five sources are used (Table 5.2); they correspond to the three levels of evaluation in Chapter 6, from controlled geometry with known truth to real subjects with tape references.

Synthetic sections. Disjoint development and test configurations of known cross-sections: ellipses, circles, superellipses, boxes, rounded boxes, asymmetric and concave sections at orientations of 15°, 55° and 80°, grid spacings of 1, 2 and

4 mm and boundary perturbations of 0, 0.5 and 2 mm RMS. The 598 attempts include direct raster sections and their two-axis silhouette intersections.

Synthetic reconstruction primitives. A $151 \times 200 \times 121$ grid at 4 mm over an 800 mm box torso (half-width 100 mm, depth 120 mm) with prescribed limbs of 40 mm vertical and sagittal extent; the silhouettes are exact projections, so the true occupancy is known for each of the eight cases.

BodyM Test-A. BodyM [18] provides real-photo silhouettes in tight-fitting clothing and A-pose with reference measurements derived from registered 3D scans. All 87 Test-A subjects were attempted with the first frontal/profile pair of each. The supplied silhouettes bypass perception, and the scan-derived references are prediction targets rather than matched tape readings. BodyM was accessed on 22 October 2026 under its Creative Commons Attribution–NonCommercial 4.0 terms.

Development photograph pairs. A small number of private photograph pairs served as complete test inputs, exercising every stage of the frozen research code from segmentation to the final measurement records. Fig. 5.1 shows the metric frame on one such silhouette pair. These photographs remain private.

Volunteer set. Ten adult volunteers (stature 164–190 cm) were photographed indoors with a smartphone and measured with a tape. Nine wore reasonably fitted clothing and one wore loose clothing.

Table 5.2: Datasets used in the project.

Dataset	Content and reference	Size	Used in
Synthetic sections	Known convex, asymmetric and concave sections; continuous true perimeter	598 attempts	Section 6.1
Synthetic primitives	Box torso with prescribed limbs; exact silhouettes and true occupancy	8 cases	Section 6.2
BodyM Test-A [18]	Real-photo binary silhouettes; scan-derived measurements	87 subjects	Section 6.3
Development pairs	Frontal and profile photographs of volunteers	3 pairs	Section 6.7
Volunteer set	Smartphone photographs; tape references	10 subjects	Sections 6.4, 6.5

5.3 Flow Of Methodology From Start To End

The methodology rests on one idea: the body measurements a tailor needs can be recovered from geometry alone, provided that the geometry is made consistent and its assumptions are stated. It proceeds through five stages, each of which turns the output of the previous one into a more explicit geometric description of the body. The system realisation of this flow, and the tier on which each stage runs, are shown in Fig. 4.4 of the Design chapter.

1. Perception. Two photographs carry far more information than measurement needs. A pretrained pose and segmentation model reduces each one to the two things that geometry can use: the outline of the body as a binary silhouette, and a small set of landmarks that locate the shoulders, hips and ankles. From this point on, colour, texture and appearance play no further role.

2. Alignment and metric scale. The two silhouettes are taken from different directions and the subject may stand slightly tilted, so they are not yet comparable. A weighted Procrustes alignment brings each view to a canonical upright pose using the landmarks, and the known stature converts the height of each silhouette from pixels into millimetres. After this stage both views share one metric coordinate frame.

3. Reconstruction. With the views in a common frame, a three-dimensional occupancy model is built by intersecting the space allowed by the frontal silhouette with that allowed by the profile silhouette. Because a plain two-view intersection gives an outstretched arm the full depth of the torso behind it, a second, profile-guided reconstruction limits the depth of each frontal column by its own vertical thickness. Both reconstructions are kept, so that the effect of this constraint remains visible.

4. Measurement. The reconstructed body is sliced horizontally at chest, waist and hip levels located from the landmarks. Since two views cannot determine the exact shape of a slice, its circumference is computed by several competing section models (a constrained ellipse, a width/depth ellipse and a convex hull), each with an explicit assumption. Shoulder length and a hip-to-ankle length are derived from the silhouettes in the same metric frame.

5. Output with provenance. Every value is reported together with the reconstruction and section model that produced it, the level at which it was taken and any reliability flags. A value that cannot be computed reliably is reported as unavailable with a reason rather than replaced by a default.

The stages are deliberately training-free: apart from the pretrained perception model, every operation is a fixed geometric rule with constants that are the same for every subject. Section 5.4 gives the equations of each stage.

5.4 Explanation Of Methodology In Detail With Equations And Formulae

5.4.1 Data acquisition and on-device preprocessing

The inputs are a frontal and an approximately orthogonal profile photograph and the stature. The full crown and feet must be visible, the arms should be approximately horizontal and separated from the torso in the frontal view, and close-fitting clothing is requested. The camera is kept level with the subject's centre at a constant distance between the two views. Let X denote lateral position, Y distance downwards from the crown and Z sagittal position, all in millimetres. A pinhole camera gives $u = f_x x_c/z_c + c_u$ and $v = f_y y_c/z_c + c_v$; replacing depth by a representative z_0 yields the weak-perspective approximation used below, in which a depth displacement Δz changes magnification by approximately $-\Delta z/z_0$. Camera intrinsics and residual yaw are not estimated, so stature calibration alone does not remove perspective error.

Perception runs in the user's browser using the MediaPipe Tasks Vision WebAssembly runtime with the heavy Pose Landmarker model [6, 7, 30], in single-image mode for one person with detection and presence thresholds of 0.5. Each photograph is decoded locally, resized to fit within 1280×1920 pixels and padded on the right to a width divisible by four. The segmentation probability map is bilinearly resampled to the image size and thresholded at $p > 0.5$; the largest eight-connected component is kept and enclosed holes smaller than 2% of the foreground are filled. A second inference on the subject crop, expanded by 15% and enlarged bicubically when its longer side is below 1024 pixels, refines the landmarks; for resize factor γ, crop size (W_r, H_r) and origin (x_0, y_0), a normalised landmark (ξ, η) maps to $(x_0 + W_r\xi/\gamma,\ y_0 + H_r\eta/\gamma)$. The shoulder, hip and ankle pairs (MediaPipe indices 11–12, 23–24, 27–28) and the elbows and wrists form the ten landmarks; a photograph is rejected on the device if more than three have visibility below 0.3.

5.4.2 Input layer

Before any geometry is computed, each received silhouette must be strictly binary, large enough and plausibly filled, the landmarks must be complete and normalised, and the stature must lie in a physically sensible range; the exact

conditions, Eq. (4.1), are part of the privacy design in Section 4.2.4. For the methodology the consequence is that every later stage can assume a clean binary mask M_v, ten landmarks with visibility weights and a valid stature H.

5.4.3 Weighted pose alignment

For image landmarks p_i, standardised targets q_i place each shoulder, hip and ankle pair horizontally while maintaining its pair distance and mean height. Visibility weights w_i are restricted to $[0.05, 1]$ and normalised. The similarity transform extends orthogonal Procrustes analysis [31]:

$$\min_{\sigma>0, R\in SO(2), t} \sum_i w_i \|\sigma R p_i + t - q_i\|^2. \tag{5.1}$$

With $\bar{p} = \sum_i w_i p_i$, $\bar{q} = \sum_i w_i q_i$ and $K = \sum_i w_i (p_i - \bar{p})(q_i - \bar{q})^T = U\Sigma V^T$, the solution is

$$\begin{aligned} S &= \mathrm{diag}(1, \det(VU^T)), \quad R = VSU^T, \\ \sigma &= \frac{\mathrm{tr}(\Sigma S)}{\sum_i w_i \|p_i - \bar{p}\|^2}, \qquad t = \bar{q} - \sigma R \bar{p}. \end{aligned} \tag{5.2}$$

Rotation beyond 35° or scale outside $[0.6, 1.6]$ causes rejection. Masks are warped with bilinear interpolation and re-thresholded at 0.5, and landmarks receive the same transform. This global alignment can leave articulation differences between the views.

5.4.4 Metric calibration

In each view $v \in \{f, s\}$, let t_v, b_v be the first and last filled rows and $L_v = b_v - t_v + 1$. With $H_m = 10H$, the scale is $s_v = H_m / L_v$ millimetres per pixel, and the horizontal origin c_v is the mean of the row midpoints (Fig. 5.1). A metric point projects as

$$\begin{aligned} (u_f, v_f) &= \big(c_f + X/s_f,\ t_f + Y/s_f\big), \\ (u_s, v_s) &= \big(c_s + Z/s_s,\ t_s + Y/s_s\big). \end{aligned} \tag{5.3}$$

Nearest-pixel lookup returns zero outside the image. Because stature is a common scale, continuous geometric measurements satisfy $\Delta C/C \simeq \Delta H/H$; fixed metric thresholds and voxel discretisation can change the discrete response.

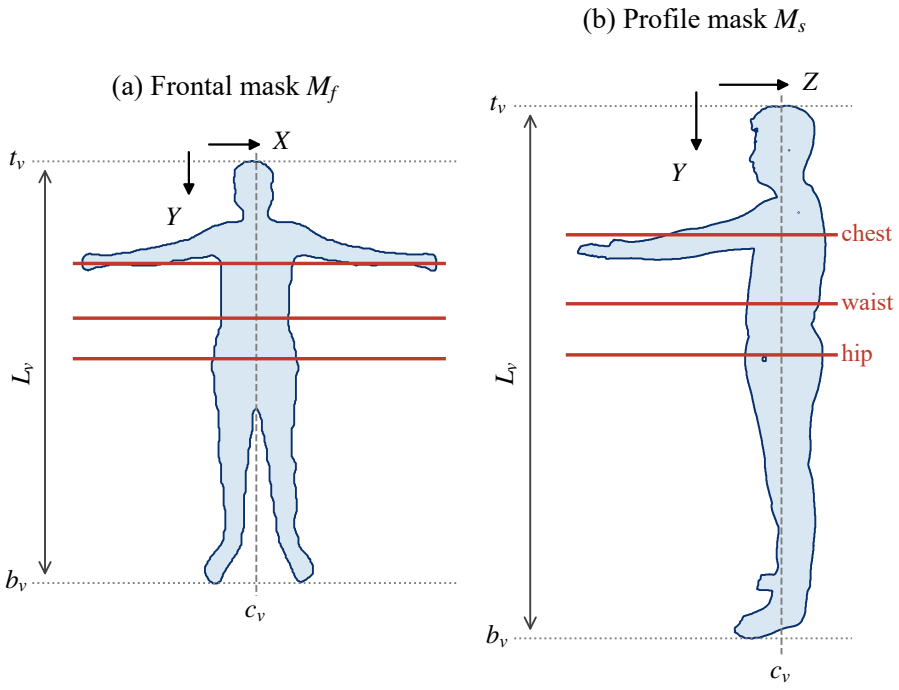


Figure 5.1: Metric coordinate frame on a development silhouette pair. X is lateral, Z sagittal and Y points down from the crown. The first and last occupied rows t_v, b_v give the pixel stature L_v and scale $s_v = H_m/L_v$; c_v is the mean row midpoint. Lines mark the chest, waist and hip levels, here derived from landmarks after alignment; in the BodyM evaluation (Section 6.3) the shoulder and hip levels are instead stature proxies $Y_s = 0.182H_m$ and $Y_h = 0.470H_m$. Binary masks of this kind are all the server receives.

5.4.5 Conventional visual hull

For binary masks M_f, M_s, the conventional orthographic occupancy is

$$O_0(X, Y, Z) = M_f(u_f, v_f)\, M_s(u_s, v_s). \tag{5.4}$$

At a given height every frontal run pairs with every profile run, so n frontal and m profile intervals yield up to nm rectangles. A horizontal limb joined to the torso forms a single frontal run and therefore inherits the full profile depth at that height. Reconstruction first samples a 10 mm grid with 30 mm margins, then resamples the occupied box at $h = 2$ mm.

5.4.6 Profile filtering

At profile row y, the widest occupied run $[\ell_y, u_y]$ has width d_y and centre c_y. Rows are rejected when

$$\begin{aligned} &d_y > 1.8 \text{ median}(d) \quad \text{or} \\ &|c_y - \text{median}(c)| > \max\{6 \text{ MAD}(c),\, 40\,\text{px}\}, \end{aligned} \tag{5.5}$$

where MAD is the unscaled median absolute deviation. The rejected mask is dilated by 51 rows; accepted endpoints are linearly interpolated, median-filtered over 101 rows and smoothed by a cubic Savitzky–Golay filter [32] of at most 61 rows, whose local coefficients solve

$$\hat{a} = (V^T V)^{-1} V^T f, \quad V_{rk} = r^k,\ k = 0, \ldots, 3, \quad \hat{f}(0) = \hat{a}_0. \tag{5.6}$$

Window lengths are in pixels, so their physical bandwidth depends on s_s. Interpolated rows are flagged because a smooth depth prior has replaced observations.

5.4.7 Profile-guided columnwise depth constraint

Converting the filtered endpoints to metric coordinates gives $\zeta_-(Y)$ and $\zeta_+(Y)$. For each frontal column X, let the vertical foreground run containing Y span $n_Y(X, Y)$ fine samples. The profile-guided reconstruction is

$$\begin{aligned} T(X,Y) &= h\, n_Y(X,Y), \qquad D_p(Y) = \zeta_+(Y) - \zeta_-(Y), \\ z_c(Y) &= \tfrac{1}{2}[\zeta_+(Y) + \zeta_-(Y)], \\ z_\pm(X,Y) &= z_c(Y) \pm \tfrac{1}{2} \min\{D_p(Y), T(X,Y)\}, \\ O_p(X,Y,Z) &= M_f(u_f, v_f)\, \mathbb{I}\{z_- \le Z \le z_+\}. \end{aligned} \tag{5.7}$$

The profile supplies a shared sagittal midpoint and depth, while the frontal vertical run bounds the thickness at the outer columns of an approximately horizontal limb. Endpoint interpolation can extend occupancy beyond the observed profile mask, so O_p is model-conditioned. Computation proceeds in 64-column chunks.

Fig. 5.2 shows the effect on a development subject, computed with both branches from the same two masks at 4 mm. In this pair the arms are raised sideways in the frontal view and held forward in the profile view, which is the worst case for Eq. (5.4). Within the arm band, the conventional hull gives the outer arm columns a mean depth of 565 mm, because every frontal arm column inherits the full profile extent of torso and forward arm; the profile-guided branch gives them 49 mm. The filter rejected 112 of 1131 profile rows and the cap acted on 6738 of

30 513 frontal columns. At the chest level, which lies inside the arm band for this subject, the torso section is 508 mm deep in O_0 and 232 mm in O_p. The constraint has known limits. For a strip of thickness t at angle φ to the horizontal, the vertical intersection is $t/|\cos\varphi|$, so the cap weakens towards vertical limbs, and a shared midpoint cannot recover a limb displaced in Z; the controlled cases of Table 6.2 quantify both effects.

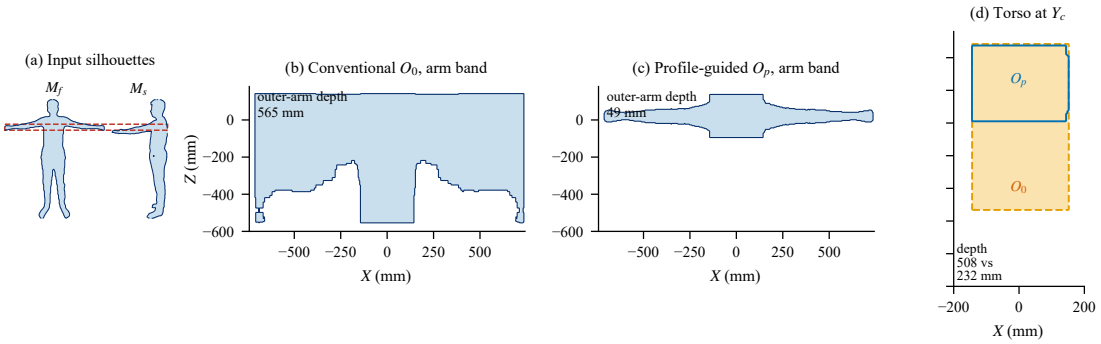


Figure 5.2: Profile-guided reconstruction on a development subject (4 mm voxels, unrounded occupancy, no display smoothing). (a) Input silhouettes; the dashed band marks the rows containing the horizontal arms. (b) Top view of the conventional hull O_0 over the arm band: each arm column inherits the full profile extent, including the forward-held arm. (c) The same view of the profile-guided reconstruction O_p of Eq. (5.7). (d) Torso section at the chest level, laterally clipped to $|X| \leq 160$ mm for display; panels (b)–(d) share the Z scale.

5.4.8 Measurement levels and waist-curvature hypothesis

A frontal torso trace follows the run nearest the preceding midpoint and is smoothed by a cubic 21-row fit. With mean shoulder and hip landmark levels Y_s, Y_h, the waist proxy is $Y_w = Y_s + 0.688(Y_h - Y_s)$, which approximates $(0.818 - 0.620)/(0.818 - 0.530)$ from engineering body-height fractions; the chest is assessed at $Y_c = Y_s + 0.35(Y_w - Y_s)$ and the hip at the landmark level. These predefined levels do not search for the maximum chest or hip girth. Each torso half-width uses its 15th percentile in a ± 40 mm neighbourhood, and both branches clip sections laterally to these bounds expanded by 1.15.

The base output applies no waist deletion. When explicitly requested, a waist-curvature hypothesis locates a local minimum of each half-width within ± 15 trace samples of Y_w, fits $w(Y) \simeq \alpha(Y - Y_w)^2 + \beta(Y - Y_w) + \gamma$ over an enclosing interval $[Y_0, Y_1]$ of length L, uses the small-slope curvature $\kappa = |2\alpha|$ and computes the

inward displacement

$$R_c = \kappa^{-1}, \quad \theta = 2\arcsin\Big(\operatorname{clip}\Big(\frac{L}{2R_c}, -0.999, 0.999\Big)\Big),$$
$$d = \min\Big\{\frac{R_c}{2}[1-\cos(\theta/2)],\ 0.40\,w_{\mathrm{loc}}\Big\}, \tag{5.8}$$

where w_{loc} is the smaller of the endpoint and local half-widths. The displacement is half the circular sagitta, with small-curvature limit $d \simeq \kappa L^2/16$. Within $Y_w \pm 15\,\mathrm{mm}$, Hann weights $\omega_j = \frac{1}{2}[1-\cos(2\pi j/(n+1))]$ taper the displacement vertically, and $t(z) = \cos[\mathrm{clip}(\pi z/(2z_h), -\pi/2, \pi/2)]$ tapers it sagittally (Fig. 5.3). Because $0 \le \omega_j t(z) \le 1$ and $d \le 0.40 w_{\mathrm{loc}}$, the hypothesis can only delete occupancy. It is skipped for prominence below 2 mm or $\kappa < 10^{-6}\,\mathrm{mm}^{-1}$, returns unavailable for invalid support, and is stored as a separate record with the base value retained. It is a geometric prior derived from the frontal width trace, not a validated anatomical correction.

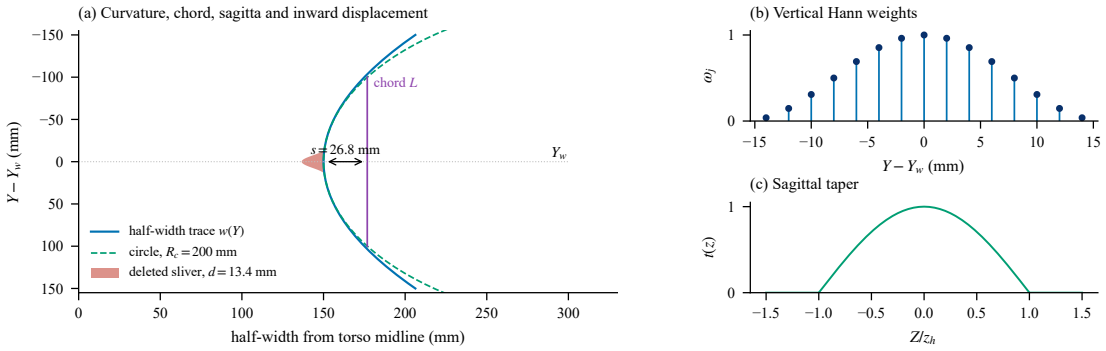


Figure 5.3: Waist-curvature hypothesis of Eq. (5.8) on a synthetic quadratic half-width trace ($\alpha = 0.0025\,\mathrm{mm}^{-1}$, $w_{\mathrm{loc}} = 150\,\mathrm{mm}$, $L = 200\,\mathrm{mm}$). (a) The curvature $\kappa = |2\alpha|$ gives $R_c = 200\,\mathrm{mm}$; the chord subtends $\theta = 60°$, the sagitta is $s = 26.8\,\mathrm{mm}$ and the inward displacement is $d = s/2 = 13.4\,\mathrm{mm}$ (small-curvature limit 12.5 mm). The shaded sliver is the occupancy deleted after the vertical taper. (b) Hann weights over $Y_w \pm 15\,\mathrm{mm}$ on a 2 mm grid. (c) Sagittal taper. The result is an optional hypothesis stored separately from the base measurement.

5.4.9 Section extraction and conic recovery

The nearest XZ slice gives a level quantisation error of at most $h/2$. Base measurement uses the unrounded, torso-clipped occupancy. Empty or multiple components, border contact and non-finite coordinates are discarded explicitly. The largest contour is transformed to metric coordinates and its convex hull is calculated with an $O(n \log n)$ monotone chain; the hull perimeter $P_H = \sum_i \|p_{i+1} - p_i\|$ is kept as an occupancy diagnostic , not as a true body circumference. For hull

vertices normalised as $v_i = (p_i - \bar{p})/\rho = (\xi_i, \eta_i)^T$, the fixed-constant conic solves

$$\min_{\beta} \|M\beta + \mathbf{1}\|^2, \qquad M_i = (\xi_i^2, \xi_i\eta_i, \eta_i^2, \xi_i, \eta_i), \tag{5.9}$$

and its centre and axes follow from

$$Q = \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix}, \quad l = \begin{pmatrix} D \\ E \end{pmatrix}, \quad \mu = -\tfrac{1}{2}Q^{-1}l, \quad F = \tfrac{1}{4}l^T Q^{-1} l - 1, \tag{5.10}$$

with $(u-\mu)^T Q (u-\mu) = F$, eigenvectors giving the orientation and F/λ_i the squared semi-axes. Because an unrestricted fit may not be an ellipse [34], the primary candidate E uses the numerically stable Halíř–Flusser reduction [33] on 256 uniform arc-length samples of the hull,

$$\begin{gathered} S_1 = D_1^T D_1, \quad S_2 = D_1^T D_2, \quad S_3 = D_2^T D_2, \quad T = -S_3^{-1} S_2^T, \\ C_1^{-1}(S_1 + S_2 T) a_1 = \lambda a_1, \quad a_2 = T a_1, \end{gathered} \tag{5.11}$$

with $D_1 = (\xi^2, \xi\eta, \eta^2)$, $D_2 = (\xi, \eta, 1)$ and the admissible eigenvector satisfying $4a_1^{(1)}a_1^{(3)} - (a_1^{(2)})^2 > 0$. Eigenvalue magnitudes must exceed $10^{-12}\|Q\|_2$. A fitted ellipse describes the section under an ellipse model; it is not a recovered anatomical cross-section and may extend beyond the occupied section.

5.4.10 Perimeter computation and estimator dependence

All ellipse candidates use the complete elliptic integral of the second kind [35],

$$P_E(a,b) = 4a\,\mathcal{E}(m), \quad m = 1 - b^2/a^2, \quad C_E = P_E/10 \text{ cm}. \tag{5.12}$$

The width/depth candidate W sets (a, b) to the sorted half-extents of the section and is conditional on an axis-aligned ellipse. Two orthogonal extents do not determine a perimeter: a box and an ellipse with equal width and depth differ in circumference. Fig. 5.4 shows the consequence on two synthetic sections rasterised at 2 mm and processed by the frozen section code. For a 15°-rotated ellipse all three estimators lie within 5.3 mm of the true perimeter; for a box-like superellipse E underestimates by 18.1 mm and W by 93.3 mm, while the hull is within 2.3 mm. Estimator choice is therefore conditional on section shape, which is why the candidates are reported side by side rather than merged.

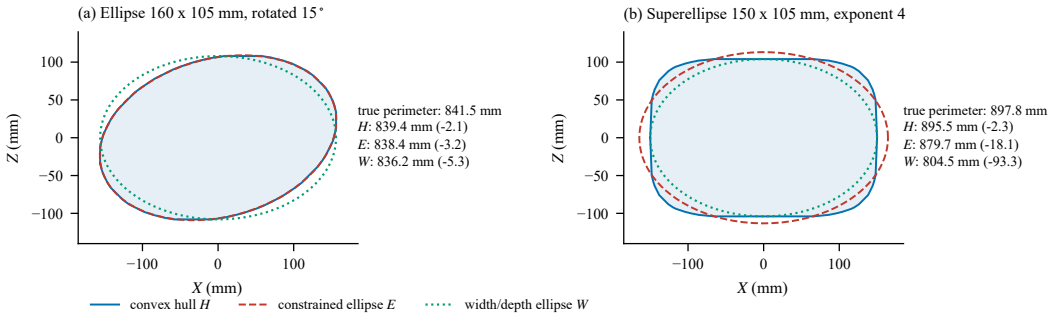


Figure 5.4: Section-model comparison on synthetic convex sections rasterised at 2 mm (cell centres). Each panel follows Sections 5.4.9 and 5.4.10: largest contour, convex hull H, 256 arc-length samples, Halíř–Flusser fit E and Eq. (5.12); W uses the section half-extents. Values in parentheses are perimeter errors (mm). (a) For a mildly rotated ellipse the candidates agree within a few millimetres. (b) For a box-like superellipse E and W underestimate the perimeter and E extends beyond the section. These illustrations are not part of the frozen evaluation of Table 6.1.

5.4.11 Shoulder and inseam models

At the acromion level $Y_a = 0.182H_m$, with unexpanded traced width W and profile depth D satisfying $D/W \leq 0.6$, the posterior shoulder path is the half-ellipse $S = \frac{1}{2}P_E(W/2, D/2)$, which assumes that the shoulder endpoints coincide with the transverse extrema of a symmetric section. Otherwise, frontal silhouette tips are located with a quadratic Savitzky–Golay trace of at most nine rows, requiring a post-peak taper of at least 5 mm, and their separation gives a projected breadth; the landmark chord is the final fallback, and the method is always recorded. The inseam output is the vertical hip-to-ankle proxy $Y_{\text{ankle}} - Y_{\text{hip}}$.

5.4.12 Measurement records and display geometry

Measurement and display use different geometry (Table 5.3). All stated perimeters come from the raw, torso-clipped occupancy and its sections. A separate display branch rounds sections, smooths the volume with a Gaussian ($\sigma_g = 1$ voxel), meshes it by marching cubes [36] and fairs the mesh for the 3D viewer; the mesh supplies no measurement. Display coordinates map (x, y, z) to $(x, z, y_{\max} - y)$, a linear map with determinant $+1$ that maintains distances. Each record includes an attempt identifier, units, source branch, level, model, fallback reason, flags and availability; unsuccessful attempts keep null values and remain in the denominator.

Table 5.3: Source geometry and defined outputs.

Output	Source and interpretation
Chest, waist, hip	Unrounded O_0 and O_p sections, each with E, W and H, reported separately
Waist hypothesis	Separate record from Eq. (5.8); base value retained
Shoulder path	Half-ellipse from frontal width and profile depth
Shoulder breadth	Silhouette tips or landmark chord, labelled by method
Hip-to-ankle proxy	Frontal landmark heights; not a tape inseam
3D viewer mesh	Display only; supplies no measurement

5.4.13 Hyperparameter and constant settings

The pipeline contains no trained parameters beyond the pretrained perception model. Its behaviour is determined by the constants in Table 5.4, which were fixed before the evaluations of Chapter 6 and are identical for all subjects.

5.4.14 Baseline models

To isolate the contribution of each component, the proposed reconstruction and section models are compared with established baselines under the same alignment, levels and torso clipping. The conventional visual hull O_0 [9] is the classical two-view intersection. The width/depth ellipse W follows the breadth-and-depth girth models of photographic anthropometry [11, 12]. The vertex-weighted conic and the historical opening comparator L represent the earlier implementation, in which sections were morphologically rounded (closing 4 mm, opening radius $0.9\min(W/2, D/2)$) before fitting. The convex-hull perimeter H is a model-free diagnostic. Within the profile-guided branch, the ablation compares the widest profile run without filtering, the filtered interval without the column cap, and the full proposed cap. Table 5.5 contrasts the models.

Table 5.4: Hyperparameter and constant settings of the proposed pipeline.

Parameter	Description	Value
On-device perception		
Pose model	MediaPipe Pose Landmarker variant	Heavy
Running mode	Images and persons per call	Single image, 1
Confidence	Detection and presence threshold	0.5
Input bound	Maximum resized photograph	1280×1920 px
Mask threshold	Segmentation probability	$p > 0.5$
Hole filling	Largest enclosed hole filled	2% of foreground
Crop margin	Landmark-refinement crop	15%
Crop upscale	Minimum longer side (bicubic)	1024 px
Visibility	Landmark visibility; max. failures	0.3; 3 of 10
Server validation and alignment		
Mask checks	Min. side; foreground fraction	200 px; 2–85%
Stature range	Accepted H	50–260 cm
Weight clip	Landmark visibility weights	$[0.05, 1]$
Rotation limit	Maximum Procrustes rotation	35°
Scale range	Accepted similarity scale	$[0.6, 1.6]$
Reconstruction		
Coarse grid	Spacing; lateral/sagittal margin	10 mm; 30 mm
Fine grid h	Voxel spacing	2 mm
Width outlier	Profile run width	$> 1.8\,\text{median}$
Centre outlier	Profile run centre	$\max\{6\,\text{MAD}, 40\}$ px
Rejection dilation	Rows around rejected rows	51
Median filter	Endpoint median window	101 rows
Savitzky–Golay	Order; maximum window	3; 61 rows
Chunk size	Columns per processing chunk	64
Membership ϵ	Numerical depth tolerance	$10^{-8}h$
Measurement		
Waist factor	$Y_w = Y_s + f(Y_h - Y_s)$	0.688
Chest factor	$Y_c = Y_s + f(Y_w - Y_s)$	0.35
Acromion level	Shoulder level from crown	$0.182H_m$
Half-width	Percentile; neighbourhood	15th; ± 40 mm
Lateral clip	Torso bound expansion	1.15
Trace smoothing	Order; window	3; 21 rows
Arc samples	Hull resampling for E	256
Eigen tolerance	Minimum $\lvert\lambda\rvert$ relative to $\lVert Q\rVert_2$	10^{-12}
Shoulder model	Applicability of half-ellipse	$D/W \leq 0.6$
Shoulder tips	SG window; min. taper; ratio	9 rows; 5 mm; $[1, 1.5]$
Waist hypothesis	Search; prominence; cap	± 15; 2 mm; $0.40w_{\text{loc}}$
Display surface (not measured)		
Smoothing	Gaussian σ_g	1 voxel
Fairing	Smoothing passes; λ; ν	12; 0.5; -0.53
Mesh size	Simplification target	150 000 faces

Table 5.5: Comparison analysis of reconstruction and section models.

Model	Working nature	Strengths	Limitations
Conventional hull O_0	Intersects back-projected frontal and profile silhouettes.	Simple; contains the observed silhouettes.	Limbs inherit profile depth; nm run pairings inflate sections.
Width/depth ellipse W	Ellipse with semi-axes equal to section half-extents.	Robust; needs only two extents; unaffected by interior shape.	Ignores section shape and rotation; box-like sections underestimated.
Vertex-weighted conic	Unconstrained algebraic fit to hull vertices.	Closed form; fast.	May return a non-ellipse; unavailable for four-vertex hulls.
Opening comparator L	Morphological rounding before fitting (historical).	Removes small protrusions.	Changes the measured contour; frequent unavailability.
Convex hull H	Polygon perimeter of the section hull.	Model-free; tracks unrounded occupancy.	Inherits every reconstruction artefact.
Constrained ellipse E	Halíř–Flusser ellipse-specific fit on arc-length samples.	Always an ellipse; numerically stable.	Sensitive to occupancy errors that W ignores.
Proposed O_p with $E/W/H$	Filtered profile interval capped by frontal column thickness.	Limits limb-depth replication; training-free; auditable.	Cap weakens for vertical or sagittally displaced limbs.

5.4.15 Performance measures

For subject i and site k, the signed error between estimate and reference is $e_{i,k} = \hat{m}_{i,k} - m_{i,k}$; positive values denote over-estimation. Over the n_k subjects for which both values exist, the following measures are reported:

$$\mathrm{MAE}_k = \frac{1}{n_k}\sum_i |e_{i,k}|, \qquad \mathrm{RMSE}_k = \Big(\frac{1}{n_k}\sum_i e_{i,k}^2\Big)^{1/2}, \tag{5.13}$$

$$\mathrm{Bias}_k = \frac{1}{n_k}\sum_i e_{i,k}, \qquad \mathrm{MAPE}_k = \frac{100}{n_k}\sum_i \frac{|e_{i,k}|}{m_{i,k}}\,\%. \tag{5.14}$$

MAE is the primary measure because it is in physical units and is not dominated by a single large error; RMSE penalises large errors and, compared with MAE, reveals outliers such as the loose-clothing case; bias exposes systematic over- or under-estimation; and MAPE normalises for body size. Agreement is examined through signed differences rather than correlation [37]. For the synthetic reconstruction

experiments the volumetric overlap with the known truth V is

$$\mathrm{IoU} = \frac{|\hat{V} \cap V|}{|\hat{V} \cup V|}. \tag{5.15}$$

Because rejection is an explicit outcome, availability $A_k = n_k/N$ is reported with every error, and no error is computed on an imputed value. Paired comparisons between configurations a and b use $\Delta\mathrm{MAE} = \mathrm{MAE}^{(b)} - \mathrm{MAE}^{(a)}$ on the subjects common to both.

Within the ellipse model, the first-order sensitivity of Eq. (5.12) is

$$\begin{aligned} \Delta P_E &\simeq P_a\,\Delta a + P_b\,\Delta b, \\ P_a &= \frac{4\,[\mathcal{E}(m) - (1-m)\,\mathcal{K}(m)]}{m}, \\ P_b &= \frac{4\,b\,[\mathcal{K}(m) - \mathcal{E}(m)]}{a\,m}, \end{aligned} \tag{5.16}$$

with $P_a = P_b = \pi$ in the circle limit [35]. Eq. (5.16) describes perturbations within the ellipse model only; it is not a covariance of the whole pipeline. Empirical mask and capture covariances are not available, and rasterisation, component changes and rejection make the pipeline non-smooth, so calibrated confidence intervals [38] are not reported. Model mismatch, such as the box-like section of Fig. 5.4, lies outside Eq. (5.16).

Chapter 6 Results and Discussion

The pipeline is evaluated at three levels of increasing realism, which are kept distinct: controlled synthetic geometry with known truth, the external BodyM benchmark with scan-derived references [18], and volunteers with tape references. Development tests of the implementation are reported separately.

6.1 Section Models on Synthetic Geometry

The conic recovery helper reproduces the axes of 27 noiseless transformed ellipses with a maximum relative error of 3.35×10^{-11}. Disjoint development and test configurations then cover ellipses, circles, superellipses, boxes, rounded boxes, asymmetric and concave sections at orientations of 15°, 55° and 80°, grid spacings of 1, 2 and 4 mm and boundary perturbations of 0, 0.5 and 2 mm RMS. The 598 attempts include direct raster sections and their two-axis silhouette intersections, and all candidates are reported without truth-based selection (Table 6.1).

Table 6.1: Test-family MAE (mm) of section models against continuous truth. E_d, E_p: constrained ellipse on direct/projected occupancy; W: width/depth ellipse; H_d: direct convex hull; L_d: historical opening comparator (unavailable attempts in parentheses).

Family	n	E_d	E_p	W	H_d	L_d
Box	27	8.11	157.16	57.35	2.67	— (27)
Circle	27	5.77	74.21	2.85	7.15	3.67 (0)
Concave	27	29.98	135.81	14.27	5.90	— (27)
Asymmetric	27	2.43	137.24	8.57	6.67	123.82 (10)
Ellipse	54	5.42	127.93	8.52	6.75	60.61 (18)
Rounded box	27	22.07	146.84	52.71	5.49	— (27)
Superellipse	54	18.78	131.67	34.59	6.55	124.27 (44)

On direct occupancy the constrained ellipse is best for elliptical and asymmetric sections, the convex hull for box-like and concave ones, and the width/depth ellipse for circles, in line with Fig. 5.4. For noiseless ellipse and circle sections the constrained-fit MAE falls from 9.37 to 4.24 and 2.54 mm as the grid is refined from 4 to 2 and 1 mm, whereas the historical opening comparator stays at 35.32, 34.04 and 39.28 mm: removing forced rounding preserves the observed contour. Projection is the dominant error source. Two orthogonal silhouettes reduce every section to its bounding rectangle, so boxes and ellipses with equal extents become indistinguishable and E_p errors exceed 74 mm in every family; the vertex-weighted

conic is unavailable for all 243 projected test hulls because they have only four vertices. Two-view girths therefore depend on a section model, and the models are reported side by side.

6.2 Profile-Guided Reconstruction

Controlled experiments use a $151 \times 200 \times 121$ grid at 4 mm over an 800 mm box torso (half-width 100 mm, depth 120 mm) with prescribed limbs of 40 mm vertical and sagittal extent; the silhouettes are exact projections. Table 6.2 reports the volumetric IoU of Eq. (5.15).

Table 6.2: Synthetic occupancy IoU against prescribed truth. H: conventional hull; W: widest profile run, no filtering; F: filtered interval, no cap; C: filtered interval with the proposed column cap.

Case	H	W	F	C
Horizontal limbs	0.94769	0.94769	0.94769	1.00000
Extra profile run	0.90682	0.94769	0.94769	1.00000
Merged artifact	0.81423	0.81423	0.94769	1.00000
True depth bulge	0.87533	0.87533	0.82112	0.86010
Displaced limb	0.90682	0.89322	0.89322	0.94107
Oblique prism	0.94769	0.94769	0.94769	1.00000
Vertical limb	0.90914	0.90914	0.90914	0.90914
Smooth depth	0.96696	0.96696	0.81606	0.85810

Selecting one profile run removes separated contamination, filtering removes the merged artifact, and the column cap removes replicated limb depth, giving IoU 1.000 for four of the eight constructed primitives and raising the displaced-limb case from 0.907 to 0.941. The table also delimits the method. When the true depth varies smoothly or bulges, the filter treats genuine variation as noise and lowers IoU, and a vertical limb is unaffected because its vertical run spans the torso. Repeating the smooth-depth case at 2, 4 and 8 mm with fixed pixel windows gives IoUs of 0.977, 0.858 and 0.810, quantifying the dependence of the filter's physical bandwidth on resolution. On real silhouettes (Fig. 5.2) the same mechanism reduces the depth assigned to the outer arm columns from 565 to 49 mm.

The waist-curvature hypothesis was tested on 32 configurations: it never adds occupancy, but at $d = 20$ mm a centred box loses 7988 voxels without any change in its convex perimeter, while shifting it 40 mm sagittally changes the perimeter

by -36.69 mm. Volume change therefore cannot stand in for a girth result, which is why the hypothesis is kept as a separate record.

6.3 External Benchmark: BodyM

BodyM [18] provides real-photo silhouettes in tight-fitting clothing and A-pose with reference measurements derived from registered 3D scans. All 87 Test-A subjects were attempted with the first frontal/profile pair of each; the method, configuration and cohort rule were frozen before reference values were read, and because the project had used the dataset before, this is a retrospective external benchmark. Supplied silhouettes bypass perception, and stature proxies $Y_s = 0.182H_m$, $Y_h = 0.470H_m$ define the levels. Thirty subjects fail the unchanged border-contact criterion and are reported as unavailable rather than imputed.

Table 6.3: BodyM Test-A prediction errors (mm). E: constrained ellipse; W: width/depth ellipse. Availability is out of 87 attempted subjects.

Site	Branch	Avail.	MAE E	RMSE E	MAE W	RMSE W
Chest	Conventional	57/87	340.60	350.09	140.99	155.10
	Profile	57/87	321.50	330.37	134.33	148.21
Waist	Conventional	55/87	93.66	121.61	58.08	72.97
	Profile	54/87	96.23	121.09	54.12	68.21
Hip	Conventional	56/87	54.24	68.45	108.42	118.58
	Profile	57/87	49.54	60.36	113.20	122.22

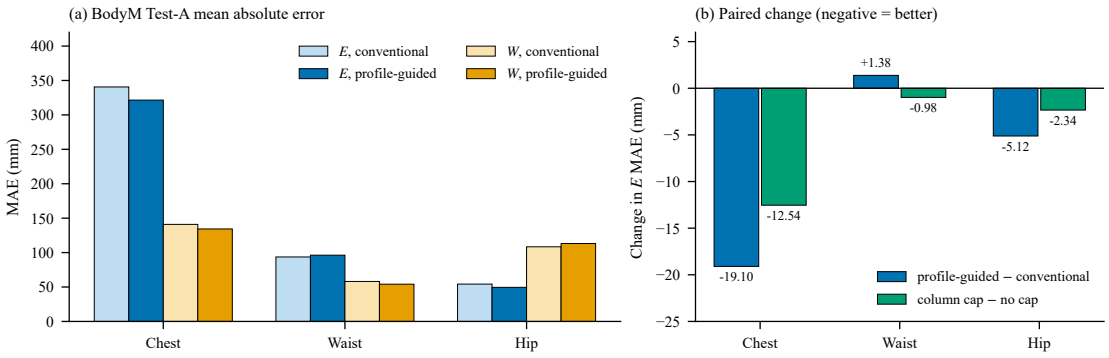


Figure 6.1: BodyM Test-A results. (a) MAE of the constrained ellipse E and width/depth ellipse W for both reconstruction branches (Table 6.3). (b) Paired change in constrained-fit MAE on common subjects: profile-guided minus conventional ($n = 57, 54, 56$) and column cap minus filtered/no-cap occupancy ($n = 57, 54, 57$).

Table 6.3 and Fig. 6.1 show that, on common subjects, the profile-guided branch improves the constrained ellipse by 19.10 mm at chest and 5.12 mm at hip

relative to the conventional hull, and changes waist by +1.38 mm. The common-grid ablation isolates the column cap: relative to the filtered interval without a cap, it reduces constrained-fit MAE by 12.54, 0.98 and 2.34 mm at chest, waist and hip; at chest 29 predictions improve and 28 are unchanged. The cap alters the clipped section in 45 of 57 chest, 4 of 57 waist and 18 of 57 hip cases, where the arms are closest to the torso, yet leaves every width/depth ellipse unchanged because the section extents do not move. The benefit of the cap thus depends on the section estimator, and the best estimator depends on the site: the profile-branch hip MAE of 49.54 mm with E is less than half that of W (113.20 mm), whereas W is better at chest and waist. The large chest errors reflect a definitional mismatch as well as reconstruction error: BodyM's scan-derived chest follows a vertex path at the fullest chest, while the pipeline evaluates a proportional proxy level with the arms close to the torso in A-pose. No estimator was selected after seeing these targets.

6.4 Volunteer Evaluation with Tape References

Ten adult volunteers (stature 164–190 cm) were photographed in ordinary indoor rooms with a smartphone and measured with a tape upright, with the hip at the fullest and the waist at the narrowest region; where a range was recorded, its midpoint is used. Repeat counts, raw readings and the chest tape level were not recorded. The deployed pipeline produced the values in Table 6.4. Nine subjects wore reasonably fitted clothing; subject S10 wore a loose polo shirt and loose trousers with a gathered elastic waistband (Fig. 6.3) and is analysed separately as a negative example, alongside the positive example S8 (Section 6.5). The inseam output is the hip-to-ankle proxy and is not compared, because the recorded leg reference measures a different length. With nine fitted subjects this is a preliminary evaluation.

Table 6.4: Volunteer results (cm). Est.: pipeline estimate; Δ: estimate minus tape reference; —: no reference recorded. S10 wore loose clothing.

		Chest		Waist		Hip		Shoulder		Inseam
Subject	Stature	Est.	Δ	Est.	Δ	Est.	Δ	Est.	Δ	Est.
S1	172.0	90.57	−1.93	83.51	−1.49	94.71	−3.29	50.53	—	67.45
S2	177.0	88.65	−0.35	94.26	+0.26	100.21	—	44.74	−3.26	62.06
S3	173.0	85.86	−8.12	83.31	−1.78	93.40	−5.03	48.81	+1.19	71.80
S4	175.2	93.92	−8.08	87.17	−4.83	102.15	−2.85	47.68	+2.68	70.13
S5	164.0	91.92	−2.58	78.78	+3.28	94.44	+0.44	46.97	−3.03	64.53
S6	176.0	100.21	−3.79	94.88	+5.88	99.57	−4.43	50.38	+0.38	71.17
S7	180.3	102.67	−1.83	95.86	+9.86	104.39	−2.61	50.83	+1.83	70.67
S8	172.0	90.36	+2.36	82.68	+0.68	97.05	+1.05	42.98	−0.02	68.49
S9	173.0	88.16	+1.16	80.78	+4.78	88.21	−2.79	46.01	+3.01	69.80
S10 (loose)	190.0	96.51	**+11.51**	85.27	**+12.27**	99.57	+3.57	50.12	+5.12	76.91

Table 6.5: Volunteer error statistics (cm) for the nine subjects in fitted clothing and for all ten subjects including the loose-clothing case.

Site	Cohort	n	MAE	RMSE	Bias	MAPE
Chest	Fitted	9	3.36	4.30	−2.57	3.46%
	All	10	4.17	5.47	−1.17	4.47%
Waist	Fitted	9	3.65	4.65	+1.85	4.32%
	All	10	4.51	5.88	+2.89	5.57%
Hip	Fitted	8	2.81	3.16	−2.44	2.81%
	All	9	2.90	3.21	−1.77	2.91%
Shoulder	Fitted	8	1.93	2.26	+0.35	4.11%
	All	9	2.28	2.73	+0.88	4.91%

For the fitted cohort (Table 6.5), MAEs are 3.36 cm at chest, 3.65 cm at waist, 2.81 cm at hip and 1.93 cm at shoulder, corresponding to MAPEs of 2.8–4.3%; 7 of 9 chest and waist estimates and 7 of 8 hip estimates fall within ±5 cm (Fig. 6.2). The signed errors are not random. Chest is underestimated on average (−2.57 cm), with the two largest errors (−8.12 and −8.08 cm) for subjects whose tape chest was probably taken at the fullest level while the pipeline uses a proportional proxy level that does not search for the maximum. Hip is likewise underestimated (−2.44 cm), consistent with the landmark hip level lying above the fullest region. Waist is overestimated (+1.85 cm), because the tape is placed at the narrowest point while the proxy level Y_w can fall away from it (S7, +9.86 cm). These level-

definition effects appear to account for much of the residual error, which suggests that searching for the extremal level is a more valuable next step than refining the section model; the small cohort does not allow this to be tested statistically.

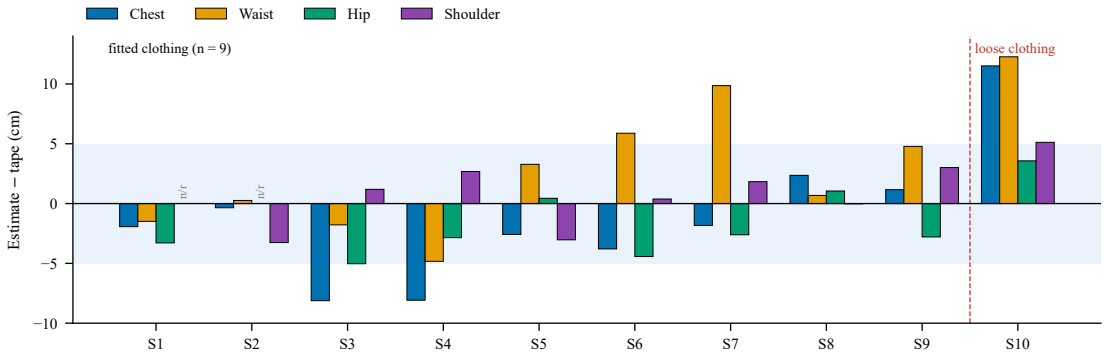


Figure 6.2: Signed errors of the volunteer evaluation (estimate minus tape) for nine subjects in fitted clothing (S1–S9) and one in loose clothing (S10). The shaded band marks ±5 cm; n/r: no reference recorded. S10 is the only subject overestimated at all four sites.

6.5 Positive and Negative Examples: Effect of Clothing

Because the silhouettes record the clothed outline, the garment worn during capture is the factor that most clearly separates good results from poor ones in the volunteer set. Two subjects at opposite ends of the error range illustrate this: S8, a positive example in fitted clothing whose estimates agree closely with the tape at every site, and S10, a negative example in loose clothing who is overestimated at every site. Table 6.6 compares them with the fitted-cohort MAE of Table 6.5.

Table 6.6: Signed errors (cm, estimate minus tape) of the positive example S8 and the negative example S10, with the MAE of the subjects in fitted clothing for reference ($n = 9$ for chest and waist, $n = 8$ for hip and shoulder).

Case	Chest	Waist	Hip	Shoulder	Mean $\lvert\Delta\rvert$
S8, fitted (positive)	+2.36	+0.68	+1.05	−0.02	1.03
S10, loose (negative)	+11.51	+12.27	+3.57	+5.12	8.12
Fitted-cohort MAE	3.36	3.65	2.81	1.93	—

6.5.1 Positive example: subject S8 in fitted clothing

Subject S8 (stature 172 cm) shows the pipeline working as intended. The estimates for S8 are 90.36 cm at chest, 82.68 cm at waist, 97.05 cm at hip and 42.98 cm at shoulder (Table 6.4), giving signed errors of +2.36, +0.68, +1.05 and −0.02 cm. All four lie well inside the ±5 cm band of Fig. 6.2, every one of them is smaller

than the corresponding fitted-cohort MAE, and the mean absolute error over the four sites, 1.03 cm, is the lowest of all volunteers.

The result is consistent with the assumptions of the method being met. In fitted clothing the silhouette follows the body closely, so the frontal width and profile depth that bound each section are close to those of the body itself, and the section models of Section 5.4.10 measure a section that resembles the anatomy. The remaining errors are of the size expected from the level definitions and the section-model assumptions discussed in Section 6.4, rather than from the clothing. The example also shows that no single site dominates: a subject who follows the capture instructions can expect all girths, not only one, to be usable for choosing a garment size.

6.5.2 Negative example: subject S10 in loose clothing

Subject S10 shows how improper clothing spoils silhouette-based measurement. As Fig. 6.3 shows, he wore a loose polo shirt that hangs from the chest and shoulders and loose trousers gathered by an elastic drawstring waistband. His errors are +11.51 cm at chest, +12.27 cm at waist, +3.57 cm at hip and +5.12 cm at shoulder (Table 6.4). He is the only subject overestimated at every site, and his chest and waist errors are 3.4 times the fitted-cohort MAE; relative to the fitted cohort's chest bias of −2.57 cm, his chest estimate is displaced by more than 14 cm. Including him raises the chest RMSE from 4.30 to 5.47 cm and the waist RMSE from 4.65 to 5.88 cm (Table 6.5).



Figure 6.3: Negative example: subject S10 in loose clothing (face obscured). (a) Frontal view: A, sleeves and shoulder seams widen the shoulders; B, the shirt drapes off the chest; C, it hangs straight with no waist taper; D, the elastic waistband is gathered. (b) Profile view: E, an air gap separates the fabric from the abdomen; F, the shirt line, not the body, sets the sagittal depth.

The mechanism follows from the method. Segmentation labels clothing as foreground, so the silhouettes, and therefore both reconstructions, describe the clothed envelope, and the hull can only bound the observed shape from outside [9]; excess fabric is never removed. If loose fabric stands off the body by a uniform δ in the frontal and profile directions, both semi-axes grow by δ and Eq. (5.16) gives

$$\Delta P \approx (P_a + P_b)\,\delta. \tag{6.1}$$

For axis ratios b/a between 0.6 and 1, $P_a + P_b$ lies between 6.18 and 2π, so the chest and waist errors correspond to an average stand-off of only $\delta \approx 1.8$–1.9 cm and 2.0 cm respectively, consistent with the gap visible between the shirt and the abdomen in the profile view. This is a within-model estimate, not a measurement of the gap. The site pattern also matches the garment. At the waist the polo hangs vertically from the chest, so the frontal trace shows no taper, the level falls on the widest fabric and the gathered waistband adds bulk; this site has the largest error. At the hip the trousers rest on the widest part of the body, the stand-off is small

and the error (+3.57 cm) is only 1.27 times the fitted MAE. The shoulder error is consistent with the sleeve and seam silhouette extending beyond the acromion. Stature is not an evident cause: S10 is the tallest subject (190 cm), but stature only sets a common scale and the other subjects show no clear size-dependent bias.

6.5.3 Lessons from the two examples

Set side by side, the two examples show that the same pipeline, with the same constants, moves from close agreement with the tape to large overestimates when only the clothing changes; S8's mean absolute error is roughly one eighth of S10's. Three conclusions follow. First, for this class of method the garment is part of the measurement; geometric refinement cannot recover the body from a silhouette that does not show it, unlike approaches that model shape under clothing [29]. Second, the capture protocol must require close-fitting clothing, tucked or removed loose tops and waistbands that do not gather, as the BodyM protocol does [18]. Third, the failure has signatures that a future version could flag: a profile depth that stays constant from chest to waist, a frontal waist-to-chest width ratio close to one, and simultaneous positive deviations at all girths.

6.6 Privacy Evaluation

For one development pair of the same subject as Fig. 5.1, the two photographs occupy 518 KB (PNG, 900×1600), whereas the two binary silhouettes occupy 19.7 KB and the landmarks and stature well under 1 KB, giving a payload ratio $r \approx 26$ in Eq. (4.2). The full-resolution smartphone originals in the volunteer set (2304×4096 JPEG) are 3.4–4.0 MB per photograph, while the transmitted silhouettes remain bounded by the 1280×1920 on-device resize. More important than size is content: in the normal workflow the server never receives a pixel that is not 0 or 255, so the face, skin, clothing colour, background and file metadata are not transmitted, and the browser-side regression tests confirm that only silhouettes and landmarks are uploaded. The derived data that remain still describe body shape and must be protected by the operator.

6.7 Computational Cost

The frozen research implementation, run end to end on two development RGB pairs on a Windows workstation, completed in 8.18 and 6.34 s with peak working sets of 2.22 and 1.80 GB; concurrent benchmark activity makes these timings illustrative. Eight injected failure tests (missing pose or segmentation, clipping,

border contact, fragments, empty input, grid budget and invalid stature) produced the expected rejection records. On the device, the perception model runs twice per photograph on the CPU; no GPU and no model training are required.

6.8 Limitations

The evaluation has limitations that bound its conclusions. The volunteer cohort is small, consists of young adults , agreement analysis with matched-plane tape readings is required [37]. The BodyM comparison targets scan-derived vertex paths that differ from the pipeline's proxy planes. Two silhouettes do not determine the body uniquely, so every girth is conditional on the reconstruction branch and the section model. Camera intrinsics, yaw and capture tolerances are not estimated, and calibrated confidence intervals are not reported. The development example of Fig. 5.2 uses a single subject whose arm pose differs between views. Finally, the privacy property protects the photographs but not the derived silhouettes, whose retention policy remains to be implemented.

Chapter 7 Conclusion and Future Work

7.1 Conclusion

This project presented a two-view pipeline for body measurement from two smartphone photographs and the stature that requires no training on measurement labels and processes the photographs on the user's device. A profile-guided columnwise depth constraint caps local depth by the vertical thickness of frontal silhouette columns, limiting the limb-depth replication of the conventional visual hull, and cross-sections are measured by competing estimators whose values carry their provenance. On constructed synthetic geometry the column cap gives IoU 1.000 for four of eight limb configurations; on the BodyM benchmark it reduces constrained-ellipse chest error by 12.54 mm in paired ablation and the profile branch gives a hip MAE of 49.54 mm, while the width/depth ellipse remains better at chest and waist; and on nine volunteers in fitted clothing the pipeline achieves MAEs of 3.36, 3.65, 2.81 and 1.93 cm at chest, waist, hip and shoulder. A subject in a loose polo shirt was overestimated by 11.51 cm at chest and 12.27 cm at waist, consistent with about 2 cm of fabric stand-off, showing that clothing, not geometry, is the dominant failure mode.

7.2 Future Work

Future work will address several limitations of the current pipeline. First, the proportional proxy levels used for the chest, waist and hip will be replaced with searches for the true extremal levels of each measurement. Second, the system will detect loose clothing from profile-depth and waist-taper signatures, so that it can warn the user before a measurement is taken. Third, agreement with matched-plane tape measurements will be evaluated on a larger and more diverse cohort and across a range of capture conditions. Finally, camera yaw and perspective will be estimated directly from the landmarks rather than assumed.

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Appendix A: Snapshots

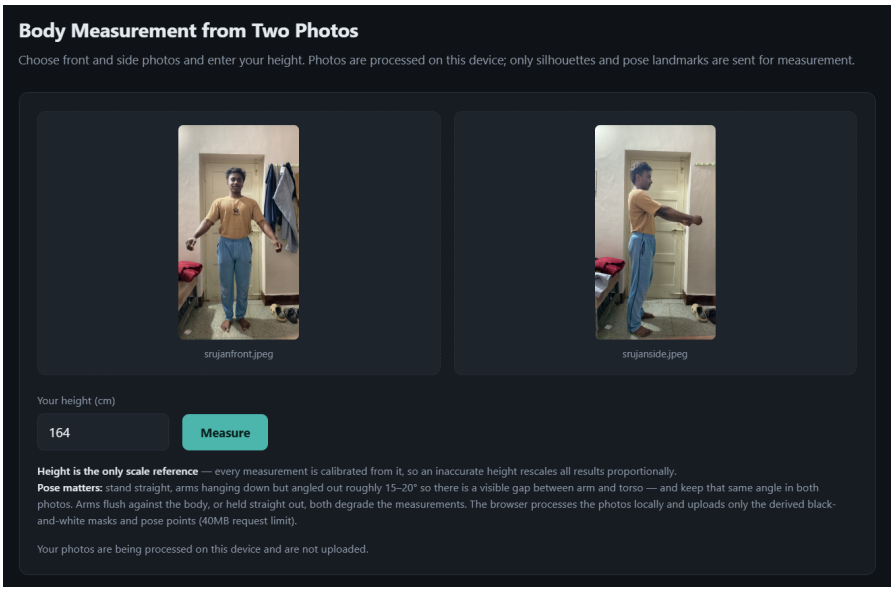


Figure A.1: Front and Side View Image Input

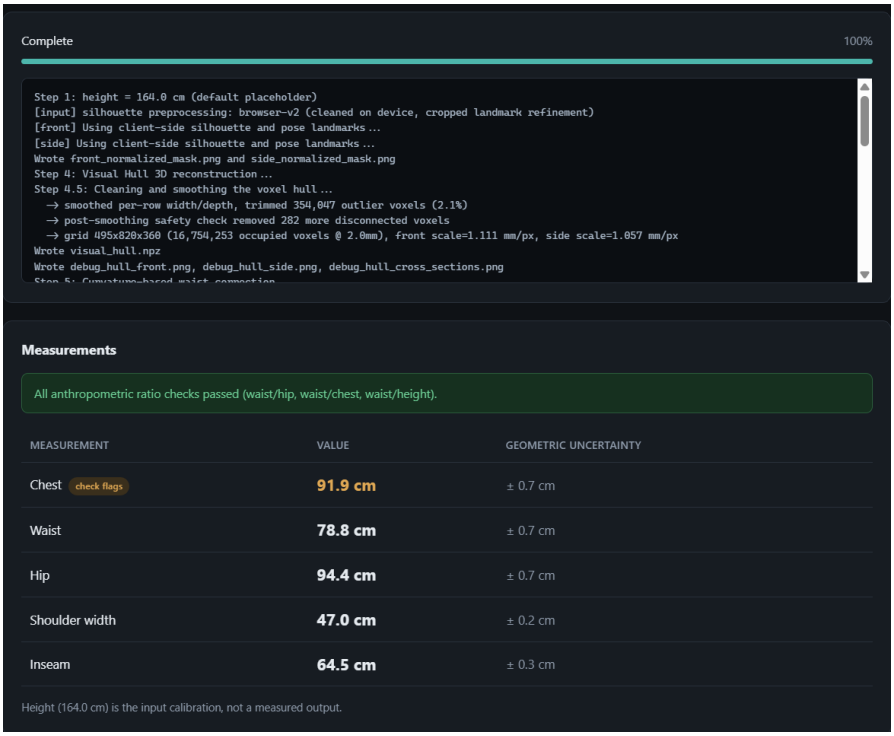


Figure A.2: Calculated Measurements



Figure A.3: Generated Visual Hull



Figure A.4: Extracted Silhouettes from Images

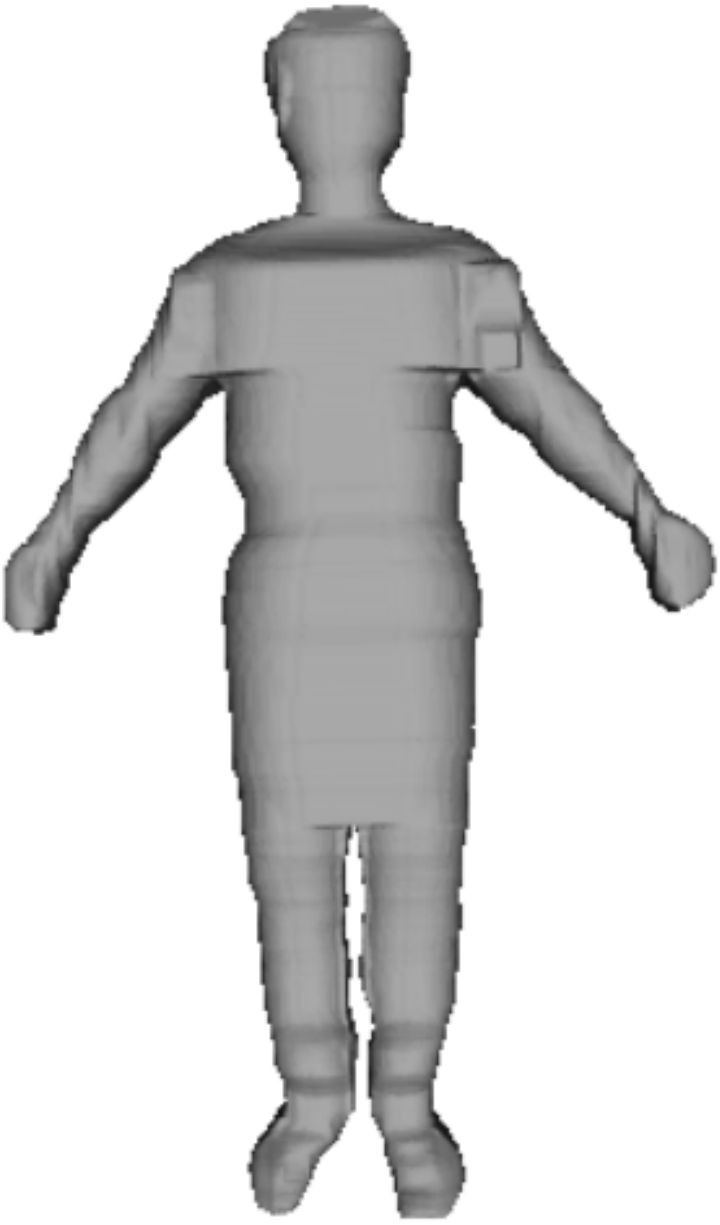


Figure A.5: 3D Model (STL) Output

Appendix B: Details of list of publications related to this project

Author Names: Dr. Jyothi S. Nayak, Sanath S. Shetty, Suhas M S, Sumanth S. Shetty, Thilak K

Paper Title: Adaptive contour grabcut : An Algorithmic Approach to Robust Human Segmentation

Name of the Conference or Journal: AIRIM (International Conference on Artificial Intelligence, Robotics, and Intelligent Manufacturing)

Place of the conference: Virtual

Date of Conference: November 5–7, 2026

Appendix C: Details of Patent

Not applicable.

Appendix D: Details of Funding

Not applicable.

Appendix E: POs and PSOs Mapped

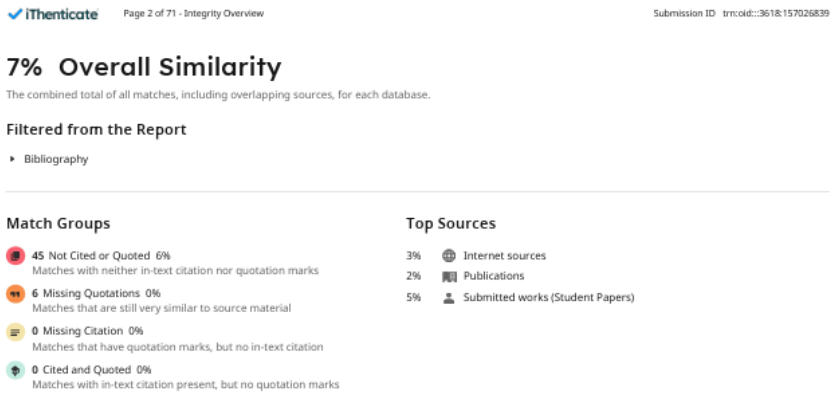
PROGRAMME OUTCOMES	Level (1/2/3)	Justification if addressed
PO1	3	Draws on linear algebra, numerical methods, computational geometry and image processing to build the measurement pipeline.
PO2	3	Analyses why two-view intersection overestimates limb depth and why a single girth hides model assumptions, and formulates the problem with explicit constraints.
PO3	3	Designs a two-tier system with on-device perception, server-side validation, two reconstruction branches and competing section estimators.
PO4	2	Investigates the method with controlled synthetic experiments, an ablation on the BodyM benchmark and a volunteer study with tape references.
PO5	3	Uses MediaPipe, WebAssembly, Python, NumPy, SciPy, OpenCV, Flask and Docker to implement and deploy the system.
PO6	1	Offers a contactless measurement route that can be used at home without a trained measurer or a body scanner.
PO7	2	Needs no GPU, no training runs and no special hardware, keeping the energy and equipment footprint of the solution small.
PO8	1	Applies data minimisation and privacy by design, and reports limitations and the loose-clothing failure openly.
PO9	2	Work on perception, reconstruction, measurement, deployment and evaluation was shared among the four team members.
PO10	2	Communicates the work through this report, a research manuscript and presentations to the department.
PO11	1	Plans the work in phases and delivers the system at no monetary cost using open-source tools and free hosting.
PO12	2	Required independent study of computer vision, 3D reconstruction, ellipse fitting and data-protection principles beyond the curriculum.

Table E.1: Programme Outcomes Mapping with Project

PROGRAMME SPECIFIC OUTCOMES	Level (1/2/3)	Justification if addressed
PSO1	3	Applies software engineering practice, including frozen configurations, regression tests and explicit rejection records, to build a dependable measurement application.
PSO2	2	Deploys a client–server web application in which inference runs in the browser and the server exposes validated submission and status endpoints.
PSO3	3	Implements efficient algorithms for alignment, visual-hull reconstruction, section fitting and perimeter computation on commodity hardware.

Table E.2: Programme Specific Outcomes Mapping with Project

Appendix F: Similarity Report, AI Generated Report


iThenticate Page 2 of 71 - Integrity Overview Submission ID trn:oid:::3618:157026839

7% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography

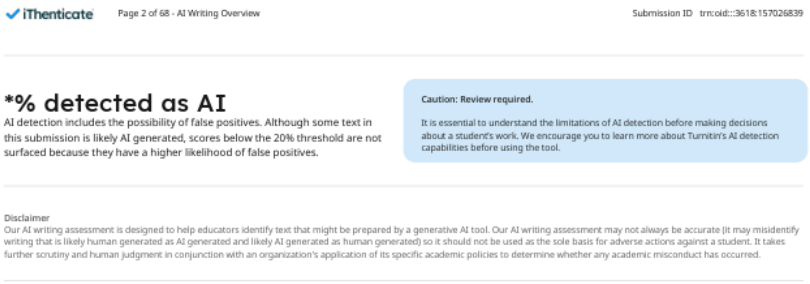
Match Groups

- 45 Not Cited or Quoted 6%
  Matches with neither in-text citation nor quotation marks
- 6 Missing Quotations 0%
  Matches that are still very similar to source material
- 0 Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

- 3% Internet sources
- 2% Publications
- 5% Submitted works (Student Papers)

Figure F.1: Similarity Detection


iThenticate Page 2 of 68 - AI Writing Overview Submission ID trn:oid:::3618:157026839

***% detected as AI**

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (it may misidentify writing that is likely human generated as AI generated and likely AI generated as human generated) so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Figure F.2: AI Writing Detection